
From Noise to Diversity: Random Embedding Injection in LLM Reasoning

Anonymous Author(s)

Affiliation

Address

email

Abstract

Recent *soft prompt* methods, which prepend or insert trainable continuous embeddings into LLM inputs as virtual tokens, optimize these embeddings to enhance mathematical reasoning. However, the mechanism by which embedding injection itself affects model behavior, independent of optimization, remains unexplored. We investigate Random Soft Prompts (RSPs), continuous vectors sampled from an isotropic Gaussian fitted to the entrywise mean and variance of the pretrained embedding table and injected without any training. Theoretically, we show that the RSP contribution at each attention layer is exactly captured by a scalar random-token attention mass whose fixed-gap upper envelope shrinks as autoregressive cache growth accumulates real tokens, and that under a local affine analysis of the perturbed logits RSPs can assign positive probability to top- K entry regions that temperature sampling cannot create by rank-preserving scaling. Empirically, RSPs increase early-stage token entropy while generation stabilizes thereafter, and combining RSPs with temperature sampling yields higher trajectory diversity as measured by Pass@ N . We further discuss the implications for GRPO training, where trajectory diversity is critical for learning signal quality.

1 Introduction

As large language models (LLMs) scale to billions of parameters, full fine-tuning becomes increasingly expensive. This has motivated parameter-efficient methods that adapt frozen models by introducing a small number of trainable parameters [Hu et al., 2022, Houlsby et al., 2019]. Among these, *soft prompts* prepend or insert learnable continuous embeddings into the input sequence, steering model behavior through the attention mechanism [Li and Liang, 2021, Lester et al., 2021]. This paradigm has been actively adopted for enhancing mathematical reasoning in LLMs [Kang et al., 2026, Xu et al., 2025, Ye et al., 2026, Hao et al., 2025, Zhang et al., 2026]. Despite their diverse designs, these methods share a common structure: they inject continuous embeddings that lie outside the pretrained vocabulary and optimize them to improve downstream performance.

However, we find that optimization is not a necessary condition for these effects. We observe that injecting randomly initialized, untrained continuous embeddings can also shift the output distribution. For instance, applying a chat template to a base model not fine-tuned with such format causes a prompt-format mismatch that degrades reasoning capability, but appending random embeddings to such mismatched inputs mitigates this degradation. If simple noise were merely perturbing the model, accuracy should worsen instead. This suggests that injecting out-of-vocabulary continuous embeddings itself, independent of any learned content, plays a non-trivial role in shaping model behavior. Existing studies primarily evaluate *soft prompt* methods through end-task accuracy, and the mechanisms by which embedding injection influences the output distribution remain insufficiently explored.

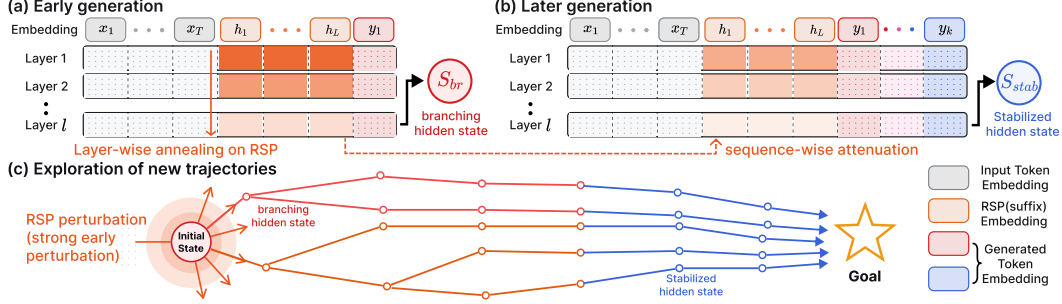


Figure 1: Conceptual overview of RSP-induced trajectory diversity (*suffix injection*). (a) Early decoding: RSP induces a branching hidden state (red) with a diverse output distribution. (b) Later decoding: cache growth attenuates the RSP attention mass, yielding a stabilized hidden state (blue). (c) As generation proceeds, the multiple trajectories opened by early RSP perturbation gradually stabilize into goal-directed completions.

In this work, we investigate the role of Random Soft Prompts (RSPs), continuous vectors sampled from an isotropic Gaussian fitted to the entrywise mean and variance of the pretrained embedding table and injected without any training. The role of randomness is not to imitate the embedding distribution; it is to ensure that repeated rollouts receive *independent* perturbations, which is what exploration requires. Theoretically, we identify an attention-level mechanism in which the RSP contribution is exactly captured by a scalar random-token attention mass whose fixed-gap upper envelope shrinks with autoregressive cache growth, and we show that under a local affine analysis of the perturbed logits RSPs can assign positive probability to top- K entry regions that temperature sampling cannot create by rank-preserving scaling. Empirically, RSPs increase early-stage token entropy while the output stabilizes in later generation stages, and combining RSPs with temperature sampling yields more diverse reasoning trajectories as measured by Pass@ N scaling. We further discuss the implications of these findings for Group Relative Policy Optimization (GRPO) [Shao et al., 2024], where trajectory diversity is essential for effective learning signals.

Our contributions are as follows:

- We give a transformer-level mechanism in which the RSP contribution is controlled by a scalar attention mass with a cache-growth upper envelope, and a distributional analysis under a local affine surrogate showing that *independent isotropic resampling*—not a fixed perturbation—can reach top- K entry regions in a way temperature sampling cannot.
- We empirically analyze how RSPs affect LLM generation, showing that RSPs increase early token entropy and produce more diverse reasoning trajectories when combined with temperature sampling, and that the accumulation gain disappears when one RSP is shared across samples.
- We discuss the implications of RSP-induced diversity for GRPO training and argue that RSP, being rank-changing while temperature is rank-preserving, provides exploration gains complementary to existing temperature-based approaches.

2 Background

2.1 Soft Prompt Methods for LLM Reasoning

Soft prompts emerged as a parameter-efficient alternative to full fine-tuning. Prefix-Tuning [Li and Liang, 2021] introduced the paradigm by prepending trainable continuous vectors to every transformer layer, Prompt Tuning [Lester et al., 2021] simplified this to input-layer-only tuning and showed that it matches full fine-tuning at sufficient scale, and P-tuning [Liu et al., 2024] extended the approach with an LSTM-based prompt encoder to model inter-position dependencies.

More recently, *soft prompts* have been actively applied to LLM reasoning. TTSV [Kang et al., 2026] optimizes prefix embeddings at test time via entropy minimization, steering the model toward high-certainty states to activate latent reasoning capabilities. SoftCoT [Xu et al., 2025] replaces

explicit chain-of-thought tokens with soft tokens generated by an assistant model, leveraging the stronger representational capacity of continuous representations over discrete language. LTPO [Ye et al., 2026] optimizes latent thought embeddings per test instance through policy gradient with a confidence-based intrinsic reward. COCONUT [Hao et al., 2025] feeds the model’s own hidden states back as input embeddings, enabling reasoning in a continuous latent space. MemGen [Zhang et al., 2026] demonstrates that embedding vectors can serve as experience memory by constructing latent token sequences that enrich the model’s reasoning. Pause tokens [Goyal et al., 2024] insert learnable tokens into the input to provide additional computation steps.

These approaches all inject out-of-vocabulary continuous embeddings optimized for task-specific objectives. Because evaluation centers on end-task accuracy, the contribution of the injection itself and that of the optimization are not separated, and the mechanisms by which such out-of-vocabulary continuous embeddings shape model behavior remain a separate and underexplored question.

2.2 Noise Injection in Neural Networks

Neural networks have incorporated noise at various stages. During training, dropout [Srivastava et al., 2014] randomly deactivates hidden units to prevent overfitting, and NEFTune [Jain et al., 2024] adds noise to token embeddings during instruction fine-tuning, reporting improvements in instruction following. These methods use noise as a regularizer and do not apply perturbations at inference time.

At inference time, perturbation-based approaches primarily target robustness and safety. Randomized smoothing [Cohen et al., 2019] injects Gaussian noise into inputs to obtain certified robustness guarantees. In the LLM setting, SmoothLLM [Robey et al., 2025] perturbs input prompts at the character level to defend against adversarial jailbreaking. These methods require multiple noisy forward passes with output aggregation, and target prediction stability rather than output diversity.

In reinforcement learning, NoisyNet [Fortunato et al., 2018] injects learnable noise into network weights to improve exploration, demonstrating that the location and structure of noise injection affect exploration quality. This motivation is closest to RSPs, which similarly target diversity, but differ in injection target (input embeddings vs. network weights) and require no learned noise parameters.

RSPs differ from these approaches in the following respects. (1) RSPs operate solely at inference without any training, whereas dropout and NEFTune inject noise during training, modifying the resulting model parameters accordingly. (2) RSPs preserve the original input and append fresh embedding vectors in a single forward pass, whereas robustness methods corrupt the input and aggregate over multiple noisy passes. (3) RSPs use purely random vectors sampled from the pretrained embedding statistics, whereas NoisyNet learns noise parameters through optimization.

3 Random Soft Prompts and Why They Work

This section states the mechanism used in the paper. A *rollout* means one full autoregressive decode under one sampled RSP. Within a rollout, an RSP enters the hidden state only through attention; the same scalar attention mass that enables an early influence is constrained by a cache-growth envelope as real tokens accumulate. Across rollouts, independent resampling can accumulate early branch differences into Pass@N gains under the task-side assumptions stated below. We keep the main statements and intuition here; further derivations are in Appendix D.

3.1 Random Soft Prompt

We start with the object being analyzed. Let $\mathbf{W}_E \in \mathbb{R}^{V \times d}$ denote the pretrained token-embedding matrix, where V is the vocabulary size and d is the hidden dimension. Write its entrywise mean and standard deviation as $\mu_E := \frac{1}{Vd} \sum_{i,k} [\mathbf{W}_E]_{i,k}$ and $\sigma_E := \text{std}(\mathbf{W}_E)$. An RSP of length L is a sequence of continuous vectors $\mathbf{H} = [h_1, \dots, h_L] \in \mathbb{R}^{L \times d}$ drawn independently from

$$h_j \sim \mathcal{N}(\mu_E \mathbf{1}_d, \sigma_E^2 \mathbf{I}_d), \quad j = 1, \dots, L. \quad (1)$$

The centered form $\bar{h}_j := h_j - \mu_E \mathbf{1}_d \sim \mathcal{N}(0, \sigma_E^2 \mathbf{I}_d)$ is a zero-mean isotropic Gaussian. The RSP can be concatenated with the input embeddings $\mathbf{X}$ at one of three positions: *prefix* ($[\mathbf{H}; \mathbf{X}]$), *infix* ($\mathbf{H}$ inserted within $\mathbf{X}$), or *suffix* ($[\mathbf{X}; \mathbf{H}]$). These positions reflect the injection strategies in prior work: TTSV [Kang et al., 2026] prepends as *prefix*, SoftCoT [Xu et al., 2025] and LTPO [Ye et al., 2026]

120 insert as *infix* between user and assistant turns, and MemGen [Zhang et al., 2026] appends as *suffix*.
 121 For repeated-sampling protocols, RSP draws and decoding randomness are independent across trials.
 122 The role of this randomness is not to imitate the embedding distribution but to give repeated rollouts
 123 *independent* perturbations, which is what exploration requires.

124 3.2 Exploration: how RSP enters the hidden state

125 We first ask how a single RSP injection perturbs the hidden state. RSP does not add to the hidden
 126 state directly; it enters only through attention weights, so its instantaneous influence within one head
 127 can collect into a single quantity: the attention mass assigned to RSP tokens.

128 Concretely, consider one attention head in self-attention layer ℓ with $n \geq 1$ unmasked real tokens,
 129 $L \geq 1$ RSP tokens, finite attention logits, attention weights $\alpha_{ij}^{(\ell)}$, and value vectors $\mathbf{v}_j^{(\ell)}, \mathbf{v}_{r_j}^{(\ell)}$ on the
 130 real and RSP sides. Define the RSP attention mass $w_{r,i}^{(\ell)} := \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)}$. Since the real-token mass
 131 is then positive, the head output decomposes *exactly* as

$$\mathbf{o}_i^{(\ell)} = (1 - w_{r,i}^{(\ell)}) \tilde{\mathbf{o}}_i^{(\ell)} + \boldsymbol{\eta}_i^{(\ell)}, \quad \tilde{\mathbf{o}}_i^{(\ell)} := \sum_{j=1}^n \frac{\alpha_{ij}^{(\ell)}}{1 - w_{r,i}^{(\ell)}} \mathbf{v}_j^{(\ell)}, \quad \boldsymbol{\eta}_i^{(\ell)} := \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)}. \quad (2)$$

132 *Derivation of Eq. (2).* Write the head output as $\sum_{j=1}^n \alpha_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} + \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)}$, then factor
 133 $1 - w_{r,i}^{(\ell)} = \sum_{j=1}^n \alpha_{ij}^{(\ell)} > 0$ out of the real-token sum. No approximation beyond the softmax-
 134 normalization identity is used. $\square$

135 The same scalar $w_{r,i}^{(\ell)}$ controls two things at once. The attenuation ratio $1 - w_{r,i}^{(\ell)}$ on the real-token
 136 signal and the upper bound $\|\boldsymbol{\eta}_i^{(\ell)}\| \leq w_{r,i}^{(\ell)} \max_j \|\mathbf{v}_{r_j}^{(\ell)}\|$ on the random contribution are both tied to
 137 it. The RSP-induced variation reduces to tracking a single number in $[0, 1]$. Early in decoding the
 138 cache is short and $w_{r,i}^{(\ell)}$ can be nontrivial, so the same identity that bounds the random contribution
 139 also lets one sampled RSP perturb early next-token decisions.

140 3.3 Annealing: KV cache growth dilutes RSP attention

141 The natural next question is how this scalar behaves as decoding proceeds. In autoregressive
 142 generation, the KV cache adds one real-token term at each step while the number of RSP terms stays
 143 fixed. Under a fixed real-vs-random logit gap, this cache growth gives an upper envelope.

144 **Theorem 1** (Attention-mass decay under cache growth). *For a query attending $n \geq 1$ real tokens
 145 and $L \geq 1$ RSP tokens with finite logits, let $\Delta_i^{(\ell)} := \min_{j \in [n]} \mathbf{q}_i^\top \mathbf{k}_j - \max_{j' \in [L]} \mathbf{q}_i^\top \mathbf{k}_{r_{j'}}$ denote the
 146 pre-scale logit gap between real and random tokens. In scaled dot-product attention,*

$$w_{r,i}^{(\ell)} \leq \frac{L}{L + n \exp(\Delta_i^{(\ell)} / \sqrt{d})},$$

147 *and the right-hand side is strictly decreasing in n and tends to 0 at fixed L and $\Delta_i^{(\ell)}$.*

148 *Proof.* Let $s_j := \mathbf{q}_i^\top \mathbf{k}_j / \sqrt{d}$, $s_{r_{j'}} := \mathbf{q}_i^\top \mathbf{k}_{r_{j'}} / \sqrt{d}$, $s_{r,\max} := \max_{j'} s_{r_{j'}}$, and $R := \sum_{j'=1}^L \exp(s_{r_{j'}})$.
 149 The gap definition gives $s_j \geq \Delta_i^{(\ell)} / \sqrt{d} + s_{r,\max}$ for every real j , and $\exp(s_{r,\max}) \geq R/L$ holds
 150 because the max dominates the average; combining the two and summing over the n real keys yields
 151 $\sum_{j=1}^n \exp(s_j) \geq (n/L) \exp(\Delta_i^{(\ell)} / \sqrt{d}) R$. Substituting into $w_{r,i}^{(\ell)} = R / (\sum_j \exp(s_j) + R)$ gives

$$w_{r,i}^{(\ell)} \leq \frac{R}{(n/L) \exp(\Delta_i^{(\ell)} / \sqrt{d}) R + R} = \frac{L}{L + n \exp(\Delta_i^{(\ell)} / \sqrt{d})}.$$

152 The derivative of the right-hand side in n is $-L \exp(\Delta_i^{(\ell)} / \sqrt{d}) / (L + n \exp(\Delta_i^{(\ell)} / \sqrt{d}))^2 < 0$, so the
 153 bound is strictly decreasing and tends to 0 as $n \rightarrow \infty$. $\square$

154 This single inequality carries two messages. The RSP length L is the design parameter we choose;
 155 it sits in the numerator and sets the *maximum* influence of the perturbation. The real-token count n
 156 grows automatically every step and sits in the denominator, so the *envelope* of the influence shrinks
 157 under the same L . What the theorem guarantees is the sequence-direction attenuation of the upper
 158 envelope at fixed (layer, query position, logit gap), not a layer-direction monotone decay; queries,
 159 keys, and hidden states co-evolve, so $\Delta_i^{(\ell)}$ also moves and the realized $w_{r,i}^{(\ell)}$ need not decrease at every
 160 step. The accurate reading is that “the envelope compatible with a fixed gap shrinks as n grows.”
 161 Because Eq. (2) bounds the random contribution by the same scalar, the perturbation magnitude
 162 inherits the same envelope. This gives an implicit explore-then-exploit interpretation of decoding
 163 without asserting stepwise monotonic decay. Figure 2 verifies the sequence-direction prediction
 164 empirically.

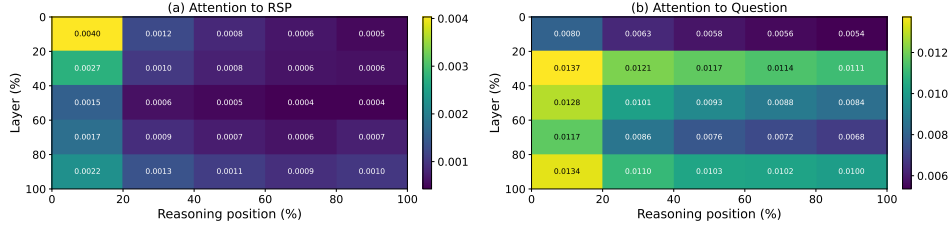


Figure 2: Per-token attention mass on Qwen2.5-Math-7B (suffix, 500 MATH-500 problems, each with an independently sampled RSP). (a) RSP-token attention decreases along the reasoning-position axis, consistent with the cache-growth bound of Theorem 1. The additional decrease along the layer axis is a separate empirical phenomenon suggesting that deeper layers sharpen the real-vs-random logit gap; it does not follow from Theorem 1 alone. (b) Question-token attention remains comparatively uniform. The reported quantity is the per-token mass of Appendix E, equal to $w_{r,i}^{(\ell)}/L$ averaged over heads and samples.

165 3.4 Performance gain: why independent reruns help

166 Theorem 1 explains what one rollout can do. The remaining question is performance: why can *fresh*
 167 RSP draws across reruns help more than repeating a single fixed perturbation? The answer needs one
 168 distributional fact about the prompt law and one separate task-side fact about productive branches.

169 The first fact is directional. The attention identity above does not require isotropy, but branch opening
 170 benefits from a perturbation law that does not under-serve whichever local direction matters for the
 171 current prompt. We state this fact for one centered RSP vector; the local surrogate below isolates one
 172 such perturbation from the length- L prompt. A minimax statement formalizes the point.

173 **Theorem 2** (Direction-agnostic minimax coverage). *Let D be any zero-mean distribution on $\mathbb{R}^d$ with*
 174 *covariance Σ_D and trace budget $\text{tr}(\Sigma_D) \leq \rho^2$. Then $\min_{\|b\|=1} \text{Var}_{h \sim D}(b^\top h) = \lambda_{\min}(\Sigma_D) \leq \rho^2/d$,*
 175 *with equality if and only if $\Sigma_D = (\rho^2/d)\mathbf{I}_d$.*

176 Equality holds for any distribution with isotropic covariance—Gaussianity is not the unique answer.
 177 The Gaussian RSP design *borrow*s the scale from the embedding distribution but discards its direc-
 178 tional geometry on purpose. The isotropic Gaussian additionally has positive density on all of $\mathbb{R}^d$
 179 (full support), so it places no prior weight on any direction (Appendix C); learned fixed prompts
 180 ($\Sigma_D = 0$), PCA-aligned noise, vocabulary-token sampling, and embedding-covariance matching are
 181 all anisotropic and so under-explore at least one direction.

182 We now formalize branch opening on the surrogate. Linearizing the logits around $\bar{h} = 0$, define
 183 the affine surrogate $z_i^{\text{lin}}(\bar{h}) := z_i + a_i^\top \bar{h}$ with $a_i := \nabla_{\bar{h}} z_i(0)$. For a baseline-excluded token i , the
 184 top- K entry region is $\mathcal{G}_{i,K} := \{\bar{h} \in \mathbb{R}^d : \#\{j \neq i : z_j^{\text{lin}}(\bar{h}) > z_i^{\text{lin}}(\bar{h})\} \leq K - 1\}$, matching the
 185 actual top- K definition.

186 **Proposition 1** (Conditional top- K entry). *If $\mathcal{G}_{i,K}$ contains a nonempty open set, the isotropic*
 187 *Gaussian’s full support assigns it positive probability, so $\mu_i := \Pr_{\bar{h}}(\bar{h} \in \mathcal{G}_{i,K}) > 0$.*

188 This proposition gives a conditional reachability statement: once such a local top- K region exists
 189 in the surrogate, the RSP law assigns it positive probability. The existence of that region remains

Table 1: Accuracy (%) across model configurations and benchmarks. RSP reports the best result across injection positions (Prefix, Infix, Suffix). Full per-position breakdown is in Appendix A. Values in parentheses indicate the difference from the baseline.

Model	MATH-500		GSM8K		AIME24	
	Baseline	RSP	Baseline	RSP	Baseline	RSP
<i>Instruct Models</i>						
LLaMA-3.1-8B-Inst	52.20	49.40 (−2.8)	85.75	86.73 (+1.0)	6.67	10.00 (+3.3)
Qwen2.5-Math-7B-Inst	83.20	84.20 (+1.0)	95.45	95.68 (+0.2)	13.33	16.67 (+3.3)
Qwen2.5-Math-1.5B-Inst	73.00	75.20 (+2.2)	85.29	85.75 (+0.5)	10.00	20.00 (+10.0)
<i>Base Models (Raw Text)</i>						
Qwen2.5-Math-7B	72.20	71.60 (−0.6)	84.15	84.31 (+0.2)	6.67	16.67 (+10.0)
Qwen2.5-Math-1.5B	61.40	64.40 (+3.0)	77.86	74.30 (−3.6)	16.67	10.00 (−6.7)
<i>Base Models + ChatML (Format Mismatch)</i>						
Qwen2.5-Math-7B	52.20	70.40 (+18.2)	58.30	75.97 (+17.7)	23.33	23.33 (+0.0)
Qwen2.5-Math-1.5B	34.40	63.40 (+29.0)	37.45	67.02 (+29.6)	13.33	20.00 (+6.7)

a task-side condition, not a consequence of the mechanism alone. To lift a single-draw opening probability to task accuracy we add two task-side assumptions: (M1) each independent RSP run reaches a correct trajectory with marginal probability at least $p_{\min}(x) > 0$, and (M2) the N runs are mutually independent.

Proposition 2 (Multi-path Pass@ N). *Under (M1)–(M2), with N independent RSP runs, $\Pr(\text{Pass}@N \text{ on } x) \geq 1 - (1 - p_{\min}(x))^N \geq 1 - \exp(-N p_{\min}(x))$.*

This division of labor is the main interpretive point. The mechanism can *admit* baseline-excluded candidates into the local candidate set when the corresponding entry region exists; whether an admitted branch is *productive* is task-dependent, which is why $p_{\min}(x) > 0$ is a separate assumption. Sharing one RSP across all samples removes this accumulation, so *independence* is essential. Section 4 tests exactly this signature.

4 Empirical Validation

We now examine how the explanation in §3 appears in actual LLM generation through three scenes. First, we ask whether RSP preserves baseline accuracy without breaking it, and recovers it in some settings (§4.2). Second, we check whether early diversification followed by late stabilization—the empirical signature suggested by the attention decomposition and cache-growth envelope—is observed (§4.3). Finally, we contrast Pass@ N between sharing one RSP across samples versus drawing a fresh RSP per sample, directly probing whether *independence* is the key (§4.4).

4.1 Experimental Setup

We evaluate on three mathematical reasoning benchmarks, MATH-500 [Lightman et al., 2024], GSM8K [Cobbe et al., 2021], and AIME24, using LLaMA-3.1-8B-Instruct [Grattafiori et al., 2024] and the Qwen2.5-Math model family [Yang et al., 2024] across instruct models, base models, and base models with mismatched chat templates—seven configurations in total. All experiments use greedy decoding to isolate the effect of RSPs from sampling stochasticity, and the answer-extraction pipeline uses fallbacks to score only the final answer. RSPs follow the definition of §3.1 and are concatenated at one of three positions (prefix, infix, suffix) with a freshly drawn $\mathbf{H}$ for each batch. Full experimental details and per-position results are in Appendix A.

4.2 How Does Untrained RSP Compare to Baselines & Optimized Soft Prompts?

We first examine how injecting random embeddings affects the model’s reasoning performance. Table 1 reports the accuracy for each model configuration, using the best RSP result across injection positions (Prefix, Infix, Suffix) and $L \in \{10, 15, 20\}$ for each benchmark.

Table 2: Comparison with existing soft prompt methods (TTSV, SoftCoT, LTPO) under unified evaluation. Baseline and RSP values are from Table 1. **Bold** denotes the best result and underline denotes the second best for each model–benchmark pair.

Model	Benchmark	Baseline	RSP	TTSV	SoftCoT	LTPO
Qwen2.5-Math-7B-Instruct	MATH-500	83.20	84.20	<u>83.80</u>	82.80	83.00
	GSM8K	<u>95.45</u>	95.68	95.30	95.00	94.84
	AIME24	13.33	<u>16.67</u>	23.30	<u>16.67</u>	13.33
Qwen2.5-Math-1.5B-Instruct	MATH-500	73.00	<u>75.20</u>	<u>75.20</u>	76.00	72.40
	GSM8K	85.29	85.75	<u>85.70</u>	84.46	84.53
	AIME24	<u>10.00</u>	20.00	6.70	6.67	<u>10.00</u>

For instruct models and base models with their native prompt formats, RSP injection maintains or slightly shifts baseline performance within a few percentage points. Most strikingly, under prompt-format mismatch — where base models receive ChatML templates they were not trained on — RSP yields pronounced improvements of up to +29 pp. These gains are difficult to attribute to simple perturbation effects, as noise that merely raises model uncertainty would further degrade performance.

We compare RSPs with prior soft prompt methods (TTSV, SoftCoT, LTPO) that optimize their embeddings for distinct objectives, under a unified evaluation protocol reported in Table 2. All methods are re-evaluated with the same unified extraction pipeline (Appendix A.2), enabling a format-agnostic comparison despite each method’s distinct prompt format and grading scheme. The results place RSPs on par with the optimized methods without requiring any training.

4.3 How Does RSP Affect the Output Distribution?

We analyze how RSP injection changes the model’s output distribution on Qwen2.5-Math-1.5B-Instruct with MATH-500, measuring per-token entropy, top-1 probability, and varentropy across the generation process. Figure 3 compares these metrics for the first 5% of generation steps across RSP variants and prior soft prompt methods (LTPO, SoftCoT, TTSV). Full statistics are in Appendix A.4.

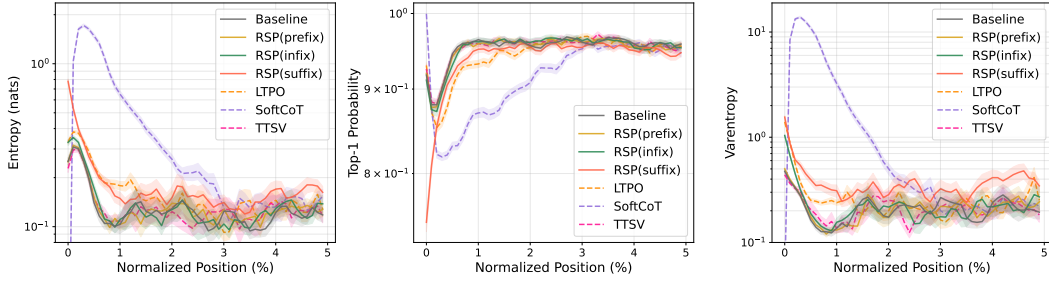


Figure 3: Mean entropy, top-1 probability, and varentropy during the first 5% of generation steps (Qwen2.5-Math-1.5B-Instruct, MATH-500). Shading indicates ± 1 SEM. Solid lines: RSP variants and the baseline. Dashed lines: prior soft prompt methods (LTPO, SoftCoT, TTSV).

These results show that RSP modifies the early-stage output distribution across all injection positions: entropy increases (the token distribution flattens), top-1 probability decreases (commitment to a dominant token weakens), and varentropy increases (indicating multimodal distributions [Ahmed et al., 2026]). High-entropy positions are known to act as critical forking points that determine reasoning trajectories [Wang et al., 2025], so this shift moves the model toward a state where multiple reasoning paths can coexist. The change is localized to the initial reasoning stage. In the middle and final portions of generation, all three metrics converge to baseline levels.

Although existing soft prompt methods are optimized for different objectives, they share a common pattern of increasing early-stage entropy. Notably, TTSV, which optimizes soft prompts using a reward that reduces the mean entropy of reasoning trajectories, still exhibits early entropy comparable to the baseline. This suggests that elevating early-stage entropy is an inherent effect of injecting non-pretrained continuous embeddings, independent of the optimization objective.

4.4 Does RSP Induce Trajectory Diversity?

Section 4.3 empirically established that RSP reshapes the early-stage output distribution. We now examine whether this shift translates into reasoning diversity at three complementary levels of analysis: token rankings, semantic content of trajectories, and task-level outcomes. The three analyses converge on a directional picture: RSP’s first-token perturbation propagates into semantically more diverse reasoning trajectories, which in turn yields greater output diversity at the task level. All three analyses share the setting: Qwen2.5-Math-1.5B-Instruct, MATH-500, suffix injection, and $L = 10$.

Token-level: distribution beyond temperature.

We quantify how much RSP shifts the first-token distribution, using temperature sampling as a baseline. Under autoregressive decoding, any trajectory-level divergence between RSP and the baseline must originate from the first generation step, a critical forking point for reasoning trajectories [Wang et al., 2025]. With the baseline at $\tau=1.0$ as reference, we compare baselines at $\tau=2.0$ and $\tau=3.0$ (16 samples per problem) against RSP at $\tau=1.0$ (averaged over 16 seeds) on four metrics: Spearman rank correlation ρ measuring whether the relative ranking of tokens is preserved, probability mass outside the baseline’s top-10 measuring support expansion to low-probability tokens, Jensen–Shannon divergence as the overall distributional distance, and Pass@1 accuracy (formal definitions in Appendix B).

Table 3: First-token distribution metrics measured against the baseline at $\tau=1.0$, comparing baselines at higher temperatures ($\tau=2.0, 3.0$) with RSP at $\tau=1.0$. RSP metrics are averaged over 16 random seeds; Acc reports mean $\pm$ std across 16 samples per problem.

Metric	$\tau=2.0$	$\tau=3.0$	RSP
Spearman ρ	1.000	1.000	0.709
Mass outside top-10	5.18%	29.11%	21.86%
JS divergence	0.060	0.218	0.131
Acc (%)	20.77 $\pm$ 1.47	1.42 $\pm$ 0.35	73.30 $\pm$ 1.07

Table 3 reports the contrast. RSP’s distributional shift falls between $\tau=2.0$ and $\tau=3.0$ in magnitude (mass outside top-10 and JS divergence), yet its mechanism and cost differ sharply. Temperature scaling is a monotone transform that preserves token ranking, so ρ relative to the baseline is exactly 1 at any τ . RSP partially preserves the ranking but induces reranking, yielding $\rho = 0.709$. The Pass@1 row exposes the cost: approaching RSP’s magnitude requires raising τ toward 3.0, but accuracy collapses to 1.42%, while RSP preserves accuracy at 73.30%. Rather than uniformly flattening the distribution as temperature scaling does, RSP produces a fundamentally different kind of perturbation.

Trajectory-level: semantic diversity.

Token-level reranking raises a second question: do first-token perturbations produce semantically diverse reasoning trajectories, or do independently perturbed trajectories converge to semantically similar ones? We randomly sample 64 problems from MATH-500 and generate 300 independent trajectories per problem under Baseline and RSP at temperature $\tau = 1.0$. Each trajectory is encoded into a dense semantic vector by BGE-M3 [Chen et al., 2024], a sentence-level multilingual embedding model, and we compute three complementary metrics over these embeddings: *pairwise cosine distance* between trajectories captures the overall semantic spread of the trajectory set; *inter-cluster distance* between centroids identified by DBSCAN [Ester et al., 1996], a density-based clustering algorithm that groups nearby trajectories without prespecifying the number of clusters, captures the separation between distinct trajectory groups; and *intra-cluster distance* captures semantic coherence within each group.

Table 4: Semantic diversity metrics (64 problems, 300 trajectories per condition).

Metric	Baseline	RSP
Pairwise cos. dist.	0.0612	0.0879
Inter-cluster dist.	0.0183	0.0982
Intra-cluster dist.	0.0526	0.0528

Table 4 reveals three concurrent patterns. Pairwise cosine distance rises by $1.4\times$, indicating that the trajectory set as a whole becomes more semantically diverse. Inter-cluster distance jumps by $5.4\times$, showing that the trajectories split into substantially more separated groups. Intra-cluster distance is essentially unchanged, indicating that semantic coherence within each group remains comparable to the baseline. RSP also yields larger pairwise distance than baseline on 56 of 64 problems (87.5%).

Outcome-level: Pass@N scaling.

Finally, we evaluate whether the token- and trajectory-level diversity translates into task-level outcome diversity through Pass@N [Chen et al., 2021], the probability that at least one of N sampled solutions is correct. We compare four conditions with $N = 16$ samples per problem on MATH-500 and AIME24: (i) *Baseline*, $\tau = 1.0$: temperature sampling only; (ii) *RSP (fixed seed)*, $\tau = 1.0$: a single RSP shared across all samples; (iii) *RSP (indep.*

302 *seed*), *greedy*: a different RSP per sample without temperature; and (iv) *RSP (indep. seed)*, $\tau = 1.0$: a
 303 different RSP per sample combined with temperature sampling.

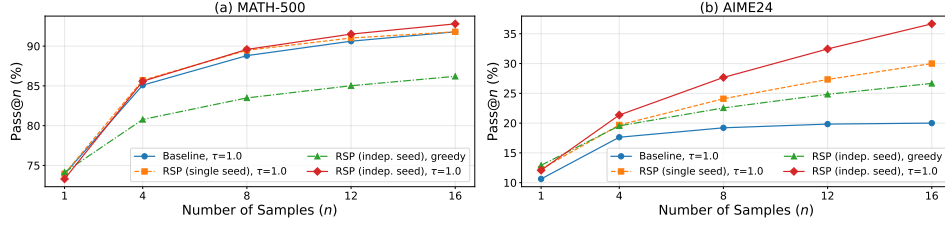


Figure 4: Pass@ N scaling on (a) MATH-500 and (b) AIME24 with Qwen2.5-Math-1.5B-Instruct, $N = 16$ samples per problem. Baseline: temperature sampling only. Fixed seed: single RSP shared across samples combined with temperature. Indep. seed: a different RSP per sample, with or without temperature.

304 Condition (iv) achieves the highest Pass@ N on both benchmarks, reaching 92.80% on MATH-500
 305 and 36.67% on AIME24 at $N = 16$. Condition (ii) shows no improvement over the baseline,
 306 suggesting that the diversity gain likely depends on varying the random embeddings across samples
 307 rather than a single fixed perturbation. Pass@1 remains comparable to the baseline on MATH-500,
 308 and on the more challenging AIME24, (iv) achieves higher Pass@1 than the baseline. Together these
 309 observations suggest that combining independent RSPs with temperature sampling may yield greater
 310 trajectory diversity while preserving single-sample accuracy.

311 5 Application: DAPO Training with RSP

312 Sections 3–4 establish what RSP does (attention-mediated branch opening) and how it behaves at
 313 inference (Pass@ N gains, rank-changing diversity unattainable by temperature). The remaining
 314 question is whether these effects transfer to training-time performance gains. Rollout diversity is a
 315 central determinant of learning in sampling-based RL, and DAPO [Yu et al., 2025] is a recent GRPO
 316 [Shao et al., 2024] variant explicitly designed to preserve that diversity at the loss level. We use it as
 317 a stringent testbed: does input-level branch opening compose with loss-level diversity preservation to
 318 yield further gains? Implementation details and full per-step results are deferred to Appendix F.

319 DAPO extends GRPO with loss-term modifications that preserve rollout diversity during long
 320 chain-of-thought RL training, shaping diversity at the loss level by reshaping how rollouts are
 321 weighted and clipped. RSP, in contrast, induces diversity at the input level by perturbing the
 322 rollout distribution before any loss is computed (Section 4.4). Let o_i be the i -th rollout response,
 323 $r_{i,t}(\theta) = \pi_\theta(o_{i,t} | q, [\mathbf{H}_g,]o_{i,<t}) / \pi_{\theta_{\text{old}}}(o_{i,t} | q, [\mathbf{H}_g,]o_{i,<t})$ the importance ratio, $\hat{A}_{i,t}$ the group-
 324 relative advantage, and $(\varepsilon_{\text{low}}, \varepsilon_{\text{high}})$ the asymmetric clip range, with the bracketed $\mathbf{H}_g$ denoting the
 325 RSP appended to the prompt. The DAPO + RSP objective is

$$\mathcal{J}_{\text{DAPO+RSP}}(\theta) = \mathbb{E} \left[\frac{1}{\sum_i |o_i|} \sum_{i,t} \min \left(r_{i,t} \hat{A}_{i,t}, \text{clip}(r_{i,t}, 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \right]. \quad (3)$$

Table 5: Per-benchmark accuracy (%) at each method’s peak step on Qwen2.5-Math-7B. **Bold** denotes the better of the two RL methods on each benchmark. Avg is the unweighted five-benchmark mean.

Method	GSM8K	MATH-500	College	Minerva	AIME 24	Avg
Base	57.2	52.6	18.3	13.6	8.6	30.06
DAPO	86.3	78.2	41.4	37.5	22.6	53.20
DAPO + RSP	87.7	78.6	42.2	39.7	23.6	54.36

326 We train Qwen2.5-Math-7B on a MATH Level-3–5 subset (~8.5K prompts) using the SimpleRL-Zoo
 327 [Zeng et al., 2025] training pipeline with the objective of Eq. (3). Each step samples $G = 8$ rollouts
 328 per prompt on a batch of 1,024 prompts under $\tau = 1.0$, followed by 4 PPO mini-batch updates of 256
 329 prompts each, for 100 training steps (about 12 epochs). We compare *DAPO* against *DAPO + RSP*,

where RSP *suffix* injection ($L = 20$ tokens) is applied during rollouts only; evaluation is performed without RSP. We evaluate on five mathematical-reasoning benchmarks (GSM8K [Cobbe et al., 2021], MATH-500 [Lightman et al., 2024], College Math, Minerva Math [Lewkowycz et al., 2022], AIME 2024). AIME 2024 is scored by Avg@32 at $\tau=1.0$ to reduce variance, while the larger benchmarks use greedy decoding. The reported average is the unweighted mean over these five benchmarks.

A first observation in Figure 5 and Table 5 is the consistent advantage of DAPO + RSP at the upper end of training: the five-benchmark average accuracy rises to 54.36% at step 90, exceeding DAPO’s own peak of 53.20% at step 80 by +1.16 pp; the gap widens at step 100 to 53.64% versus 50.54% (+3.10 pp), and the post-step-80 decline observed under DAPO alone is delayed when RSP is combined. This pattern follows from the fact that RSP and DAPO’s loss-term modifications act at separated stages of the optimization. DAPO’s modifications take already-generated rollouts and logits as input and preserve diversity at the loss level, whereas RSP modifies the hidden states from which those rollouts are produced (Section 3.2). The additional gain observed on top of DAPO’s aggressive loss-level diversity preservation suggests that hidden-state-level perturbation operates on a surface that loss-level preservation does not reach.

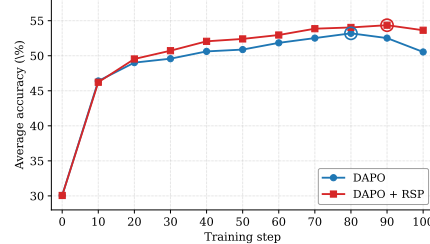


Figure 5: Five-benchmark average accuracy on Qwen2.5-Math-7B. DAPO + RSP reaches a higher peak (step 90) and stays stable through step 100.

A second observation is that the two curves cross around step 20, with DAPO + RSP showing slower reward growth in the earlier phase. This crossing is consistent with the multi-path argument of Section 3.4 reproduced in training dynamics, where deviations from the trajectories the policy would otherwise have selected expose the learning signal to a broader trajectory distribution. The cumulative effect of that broader exposure manifests as the later-stage ceiling improvement and delayed collapse. The same pattern has long been documented as the exploration–exploitation dynamics of reinforcement learning [Sutton and Barto, 2018]; the present result shows that this dynamic can also be induced from the input side through hidden-state-level perturbation.

6 Conclusion

We studied Random Soft Prompts (RSPs), isotropic Gaussian vectors fitted to the entrywise mean and variance of the pretrained embedding table and injected without training. Theoretically, the RSP contribution at each attention layer is exactly captured by a scalar attention mass whose fixed-gap upper envelope shrinks under cache growth, and under a local affine analysis RSPs can reach top- K entry regions that rank-preserving temperature scaling cannot create (§3). Empirically, RSPs raise early-stage entropy while later generation stabilizes, and the accumulation gain disappears when one RSP is shared across samples (§4). Because RSP is rank-changing while temperature is rank-preserving, the two are complementary, suggesting a sampling-level remedy for entropy collapse in GRPO [Shao et al., 2024] beyond DAPO [Yu et al., 2025] and CE-GPPO [Su et al., 2026]; training-time validation, extension beyond mathematical reasoning, and a principled choice of RSP length and injection position—together with the task-side bound $p_{\min}(x) > 0$ of Proposition 2—are natural next steps.

References

- Farhan Ahmed, Yuya Jeremy Ong, and Chad DeLuca. LogitScope: A Framework for Analyzing LLM Uncertainty through Information Metrics. In *ICLR Workshop*, 2026.
- Nora Belrose, Igor Ostrovsky, Lev McKinney, Zach Furman, Logan Smith, Danny Halawi, Stella Biderman, and Jacob Steinhardt. Eliciting Latent Predictions from Transformers with the Tuned Lens. *arXiv:2303.08112*, 2023.
- Jianlyu Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. M3-Embedding: Multi-Linguality, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation. In *Findings of ACL*, 2024.

381 Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared
382 Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, et al. Evaluating Large
383 Language Models Trained on Code. *arXiv:2107.03374*, 2021.

384 Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser,
385 Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John
386 Schulman. Training Verifiers to Solve Math Word Problems. *arXiv:2110.14168*, 2021.

387 Jeremy M. Cohen, Elan Rosenfeld, and J. Zico Kolter. Certified Adversarial Robustness via Random-
388 ized Smoothing. In *ICML*, 2019.

389 Martin Ester, Hans-Peter Kriegel, Jörg Sander, and Xiaowei Xu. A Density-Based Algorithm for
390 Discovering Clusters in Large Spatial Databases with Noise. In *KDD*, 1996.

391 Meire Fortunato, Mohammad Gheshlaghi Azar, Bilal Piot, Jacob Menick, Ian Osband, Alex Graves,
392 Vlad Mnih, Remi Munos, Demis Hassabis, Olivier Pietquin, Charles Blundell, and Shane Legg.
393 Noisy Networks for Exploration. In *ICLR*, 2018.

394 Mor Geva, Roei Schuster, Jonathan Berant, and Omer Levy. Transformer Feed-Forward Layers Are
395 Key-Value Memories. In *EMNLP*, 2021.

396 Sachin Goyal, Ziwei Ji, Ankit Singh Rawat, Aditya Krishna Menon, Sanjiv Kumar, and Vaishnavh
397 Nagarajan. Think before You Speak: Training Language Models with Pause Tokens. In *ICLR*,
398 2024.

399 Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad
400 Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela
401 Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem
402 Korenev, Arthur Hinsvark, Arun Rao, Aston Zhang, Aurelien Rodriguez, Austen Gregerson, Ava
403 Spataru, Baptiste Roziere, Bethany Biron, Binh Tang, Bobbie Chern, Charlotte Caucheteux, Chaya
404 Nayak, Chloe Bi, Chris Marra, Chris McConnell, Christian Keller, Christophe Touret, Chunyang
405 Wu, Corinne Wong, Cristian Canton Ferrer, Cyrus Nikolaidis, Damien Allonsius, Daniel Song,
406 Danielle Pintz, Danny Livshits, Danny Wyatt, David Esiobu, Dhruv Choudhary, Dhruv Mahajan,
407 Diego Garcia-Olano, Diego Perino, Dieuwke Hupkes, Egor Lakomkin, Ehab AlBadawy, Elina
408 Lobanova, Emily Dinan, Eric Michael Smith, Filip Radenovic, Francisco Guzmán, Frank Zhang,
409 Gabriel Synnaeve, Gabrielle Lee, Georgia Lewis Anderson, Govind Thattai, Graeme Nail, Gregoire
410 Mialon, Guan Pang, Guillem Cucurell, Hailey Nguyen, Hannah Korevaar, Hu Xu, Hugo Touvron,
411 Iliyan Zarov, Imanol Arrieta Ibarra, Isabel Kloumann, Ishan Misra, Ivan Evtimov, Jack Zhang,
412 Jade Copet, Jaewon Lee, Jan Geffert, Jana Vranes, Jason Park, Jay Mahadeokar, Jeet Shah, Jelmer
413 van der Linde, Jennifer Billock, Jenny Hong, Jenya Lee, Jeremy Fu, Jianfeng Chi, Jianyu Huang,
414 Jiawen Liu, Jie Wang, Jiecao Yu, Joanna Bitton, Joe Spisak, Jongsoo Park, Joseph Rocca, Joshua
415 Johnstun, Joshua Saxe, Junteng Jia, Kalyan Vasuden Alwala, Karthik Prasad, Kartikeya Upasani,
416 Kate Plawiak, Ke Li, Kenneth Heafield, Kevin Stone, Khalid El-Arini, Krithika Iyer, Kshitiz
417 Malik, Kuenley Chiu, Kunal Bhatta, Kushal Lakhotia, Lauren Rantala-Yearly, Laurens van der
418 Maaten, Lawrence Chen, Liang Tan, Liz Jenkins, Louis Martin, Lovish Madaan, Lubo Malo,
419 Lukas Blecher, Lukas Landzaat, Luke de Oliveira, Madeline Muzzi, Mahesh Pasupuleti, Mannat
420 Singh, Manohar Paluri, Marcin Kardas, Maria Tsimpoukelli, Mathew Oldham, Mathieu Rita, Maya
421 Pavlova, Melanie Kambadur, Mike Lewis, Min Si, Mitesh Kumar Singh, Mona Hassan, Naman
422 Goyal, Narjes Torabi, Nikolay Bashlykov, Nikolay Bogoychev, Niladri Chatterji, Ning Zhang,
423 Olivier Duchenne, Onur Çelebi, Patrick Alrassy, Pengchuan Zhang, Pengwei Li, Petar Vasic,
424 Peter Weng, Prajjwal Bhargava, Pratik Dubal, Praveen Krishnan, Punit Singh Koura, Puxin Xu,
425 Qing He, Qingxiao Dong, Ragavan Srinivasan, Raj Ganapathy, Ramon Calderer, Ricardo Silveira
426 Cabral, Robert Stojnic, Roberta Raileanu, Rohan Maheswari, Rohit Girdhar, Rohit Patel, Romain
427 Sauvestre, Ronnie Polidoro, Roshan Sumbaly, Ross Taylor, Ruan Silva, Rui Hou, Rui Wang, Saghar
428 Hosseini, Sahana Chennabasappa, Sanjay Singh, Sean Bell, Seohyun Sonia Kim, Sergey Edunov,
429 Shaoliang Nie, Sharan Narang, Sharath Raparthy, Sheng Shen, Shengye Wan, Shruti Bhosale,
430 Shun Zhang, Simon Vandenhende, Soumya Batra, Spencer Whitman, Sten Sootla, Stephane
431 Collot, Suchin Gururangan, Sydney Borodinsky, Tamar Herman, Tara Fowler, Tarek Sheasha,
432 Thomas Georgiou, Thomas Scialom, Tobias Speckbacher, Todor Mihaylov, Tong Xiao, Ujjwal
433 Karn, Vedanuj Goswami, Vibhor Gupta, Vignesh Ramanathan, Viktor Kerkez, Vincent Gouget,
434 Virginie Do, Vish Vogeti, Vitor Albiero, Vladan Petrovic, Weiwei Chu, Wenhan Xiong, Wenyin

435 Fu, Whitney Meers, Xavier Martinet, Xiaodong Wang, Xiaofang Wang, Xiaoqing Ellen Tan, Xide
 436 Xia, Xinfeng Xie, Xuchao Jia, Xuwei Wang, Yaelle Goldschlag, Yashesh Gaur, Yasmine Babaei,
 437 Yi Wen, Yiwen Song, Yuchen Zhang, Yue Li, Yuning Mao, Zacharie Delpierre Coudert, Zheng Yan,
 438 Zhengxing Chen, Zoe Papakipos, Aaditya Singh, Aayushi Srivastava, Abha Jain, Adam Kelsey,
 439 Adam Shajnfeld, Adithya Gangidi, Adolfo Victoria, Ahuva Goldstand, Ajay Menon, Ajay Sharma,
 440 Alex Boesenber, Alexei Baevski, Allie Feinstein, Amanda Kallet, Amit Sangani, Amos Teo,
 441 Anam Yunus, Andrei Lupu, Andres Alvarado, Andrew Caples, Andrew Gu, Andrew Ho, Andrew
 442 Poulton, Andrew Ryan, Ankit Ramchandani, Annie Dong, Annie Franco, Anuj Goyal, Aparajita
 443 Saraf, Arkabandhu Chowdhury, Ashley Gabriel, Ashwin Bharambe, Assaf Eisenman, Azadeh
 444 Yazdan, Beau James, Ben Maurer, Benjamin Leonhardi, Bernie Huang, Beth Loyd, Beto De Paola,
 445 Bhargavi Paranjape, Bing Liu, Bo Wu, Boyu Ni, Braden Hancock, Bram Wasti, Brandon Spence,
 446 Brani Stojkovic, Brian Gamido, Britt Montalvo, Carl Parker, Carly Burton, Catalina Mejia, Ce Liu,
 447 Changhan Wang, Changkyu Kim, Chao Zhou, Chester Hu, Ching-Hsiang Chu, Chris Cai, Chris
 448 Tindal, Christoph Feichtenhofer, Cynthia Gao, Damon Civin, Dana Beaty, Daniel Kreymer, Daniel
 449 Li, David Adkins, David Xu, Davide Testuggine, Delia David, Devi Parikh, Diana Liskovich,
 450 Didem Foss, Dingkan Wang, Duc Le, Dustin Holland, Edward Dowling, Eissa Jamil, Elaine
 451 Montgomery, Eleonora Presani, Emily Hahn, Emily Wood, Eric-Tuan Le, Erik Brinkman, Esteban
 452 Arcaute, Evan Dunbar, Evan Smothers, Fei Sun, Felix Kreuk, Feng Tian, Filippas Kokkinos, Firat
 453 Ozgenel, Francesco Caggioni, Frank Kanayet, Frank Seide, Gabriela Medina Florez, Gabriella
 454 Schwarz, Gada Badeer, Georgia Swee, Gil Halpern, Grant Herman, Grigory Sizov, Guangyi Zhang,
 455 Guna Lakshminarayanan, Hakan Inan, Hamid Shojanazeri, Han Zou, Hannah Wang, Hanwen Zha,
 456 Haroun Habeeb, Harrison Rudolph, Helen Suk, Henry Aspegren, Hunter Goldman, Hongyuan
 457 Zhan, Ibrahim Damlaj, Igor Molybog, Igor Tufanov, Ilias Leontiadis, Irina-Elena Veliche, Itai
 458 Gat, Jake Weissman, James Geboski, James Kohli, Janice Lam, Japhet Asher, Jean-Baptiste Gaya,
 459 Jeff Marcus, Jeff Tang, Jennifer Chan, Jenny Zhen, Jeremy Reizenstein, Jeremy Teboul, Jessica
 460 Zhong, Jian Jin, Jingyi Yang, Joe Cummings, Jon Carvill, Jon Shepard, Jonathan McPhie, Jonathan
 461 Torres, Josh Ginsburg, Junjie Wang, Kai Wu, Kam Hou U, Karan Saxena, Kartikay Khandelwal,
 462 Katayoun Zand, Kathy Matosich, Kaushik Veeraraghavan, Kelly Michelena, Keqian Li, Kiran
 463 Jagadeesh, Kun Huang, Kunal Chawla, Kyle Huang, Lailin Chen, Lakshya Garg, Lavender A,
 464 Leandro Silva, Lee Bell, Lei Zhang, Liangpeng Guo, Licheng Yu, Liron Moshkovich, Luca
 465 Wehrstedt, Madian Khabza, Manav Avalani, Manish Bhatt, Martynas Mankus, Matan Hasson,
 466 Matthew Lennie, Matthias Reso, Maxim Groshev, Maxim Naumov, Maya Lathi, Meghan Keneally,
 467 Miao Liu, Michael L. Seltzer, Michal Valko, Michelle Restrepo, Mihir Patel, Mik Vyatskov,
 468 Mikayel Samvelyan, Mike Clark, Mike Macey, Mike Wang, Miquel Jubert Hermoso, Mo Metanat,
 469 Mohammad Rastegari, Munish Bansal, Nandhini Santhanam, Natascha Parks, Natasha White,
 470 Navyata Bawa, Nayan Singhal, Nick Egebo, Nicolas Usunier, Nikhil Mehta, Nikolay Pavlovich
 471 Laptev, Ning Dong, Norman Cheng, Oleg Chernoguz, Olivia Hart, Omkar Salpekar, Ozlem
 472 Kalinli, Parkin Kent, Parth Parekh, Paul Saab, Pavan Balaji, Pedro Rittner, Philip Bontrager,
 473 Pierre Roux, Piotr Dollar, Polina Zvyagina, Prashant Ratanchandani, Pritish Yuvraj, Qian Liang,
 474 Rachad Alao, Rachel Rodriguez, Rafi Ayub, Raghotham Murthy, Raghu Nayani, Rahul Mitra,
 475 Rangaprabhu Parthasarathy, Raymond Li, Rebekkah Hogan, Robin Battey, Rocky Wang, Russ
 476 Howes, Ruty Rinott, Sachin Mehta, Sachin Siby, Sai Jayesh Bondu, Samyak Datta, Sara Chugh,
 477 Sara Hunt, Sargun Dhillon, Sasha Sidorov, Satadru Pan, Saurabh Mahajan, Saurabh Verma, Seiji
 478 Yamamoto, Sharadh Ramaswamy, Shaun Lindsay, Shaun Lindsay, Sheng Feng, Shenghao Lin,
 479 Shengxin Cindy Zha, Shishir Patil, Shiva Shankar, Shuqiang Zhang, Shuqiang Zhang, Sinong Wang,
 480 Sneha Agarwal, Soji Sajuyigbe, Soumith Chintala, Stephanie Max, Stephen Chen, Steve Kehoe,
 481 Steve Satterfield, Sudarshan Govindaprasad, Sumit Gupta, Summer Deng, Sungmin Cho, Sunny
 482 Virk, Suraj Subramanian, Sy Choudhury, Sydney Goldman, Tal Remez, Tamar Glaser, Tamara
 483 Best, Thilo Koehler, Thomas Robinson, Tianhe Li, Tianjun Zhang, Tim Matthews, Timothy Chou,
 484 Tzook Shaked, Varun Vontimitta, Victoria Ajayi, Victoria Montanez, Vijai Mohan, Vinay Satish
 485 Kumar, Vishal Mangla, Vlad Ionescu, Vlad Poenaru, Vlad Tiberiu Mihailescu, Vladimir Ivanov,
 486 Wei Li, Wenchen Wang, Wenwen Jiang, Wes Bouaziz, Will Constable, Xiaocheng Tang, Xiaojian
 487 Wu, Xiaolan Wang, Xilun Wu, Xinbo Gao, Yaniv Kleinman, Yanjun Chen, Ye Hu, Ye Jia, Ye Qi,
 488 Yenda Li, Yilin Zhang, Ying Zhang, Yossi Adi, Youngjin Nam, Yu Wang, Yu Zhao, Yuchen Hao,
 489 Yundi Qian, Yunlu Li, Yuzi He, Zach Rait, Zachary DeVito, Zef Rosnbrick, Zhaoduo Wen, Zhenyu
 490 Yang, Zhiwei Zhao, and Zhiyu Ma. The Llama 3 Herd of Models. *arXiv:2407.21783*, 2024.

491 Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason Weston, and Yuandong Tian.
 492 Training Large Language Models to Reason in a Continuous Latent Space. In *ICLR*, 2025.

493 Chaoqun He, Renjie Luo, Yuzhuo Bai, Shengding Hu, Zhen Leng Thai, Junhao Shen, Jinyi Hu,
494 Xu Han, Yujie Huang, Yankai Zhang, Jie Liu, Lei Qi, Zhiyuan Liu, and Maosong Sun. Olympiad-
495 Bench: A Challenging Benchmark for Promoting AGI with Olympiad-Level Bilingual Multimodal
496 Scientific Problems. In *Findings of ACL*, 2024.

497 Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin de Laroussilhe, Andrea
498 Gesmundo, Mona Attariyan, and Sylvain Gelly. Parameter-Efficient Transfer Learning for NLP. In
499 *ICML*, 2019.

500 Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang,
501 and Weizhu Chen. LoRA: Low-Rank Adaptation of Large Language Models. In *ICLR*, 2022.

502 Neel Jain, Ping-yeh Chiang, Yuxin Wen, John Kirchenbauer, Hong-Min Chu, Gowthami Somepalli,
503 Brian R. Bartoldson, Bhavya Kailkhura, Avi Schwarzschild, Aniruddha Saha, Micah Goldblum,
504 Jonas Geiping, and Tom Goldstein. NEFTune: Noisy Embeddings Improve Instruction Finetuning.
505 In *ICLR*, 2024.

506 Xinyue Kang, Diwei Shi, and Li Chen. Model Whisper: Steering Vectors Unlock Large Language
507 Models’ Potential in Test-time. In *AAAI*, 2026.

508 Brian Lester, Rami Al-Rfou, and Noah Constant. The Power of Scale for Parameter-Efficient Prompt
509 Tuning. In *EMNLP*, 2021.

510 Aitor Lewkowycz, Anders Andreassen, David Dohan, Ethan Dyer, Henryk Michalewski, Vinay
511 Ramasesh, Ambrose Slone, Cem Anil, Imanol Schlag, Theo Gutman-Solo, Yuhuai Wu, Behnam
512 Neyshabur, Guy Gur-Ari, and Vedant Misra. Solving Quantitative Reasoning Problems with
513 Language Models. *arXiv:2206.14858*, 2022.

514 Xiang Lisa Li and Percy Liang. Prefix-Tuning: Optimizing Continuous Prompts for Generation. In
515 *ACL-IJCNLP*, 2021.

516 Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike,
517 John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s Verify Step by Step. In *ICLR*, 2024.

518 Xiao Liu, Yanan Zheng, Zhengxiao Du, Ming Ding, Yujie Qian, Zhilin Yang, and Jie Tang. GPT
519 Understands, Too. *AI Open*, 5:208–215, 2024.

520 Aman Madaan and Amir Yazdanbakhsh. Text and Patterns: For Effective Chain of Thought, It Takes
521 Two to Tango. *arXiv:2209.07686*, 2022.

522 Alexander Robey, Eric Wong, Hamed Hassani, and George J. Pappas. SmoothLLM: Defending Large
523 Language Models Against Jailbreaking Attacks. *Transactions on Machine Learning Research*,
524 2025.

525 Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang,
526 Mingchuan Zhang, Y.K. Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the Limits of
527 Mathematical Reasoning in Open Language Models. *arXiv:2402.03300*, 2024.

528 Guangming Sheng, Chi Zhang, Zilingfeng Ye, Xibin Wu, Wang Zhang, Ru Zhang, Yanghua
529 Peng, Haibin Lin, and Chuan Wu. HybridFlow: A Flexible and Efficient RLHF Framework.
530 *arXiv:2409.19256*, 2024.

531 Nitish Srivastava, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov.
532 Dropout: A Simple Way to Prevent Neural Networks from Overfitting. *Journal of Machine
533 Learning Research*, 15:1929–1958, 2014.

534 Zhenpeng Su, Leiyu Pan, Minxuan Lv, Yuntao Li, Wenping Hu, Fuzheng Zhang, Kun Gai, and Guorui
535 Zhou. CE-GPPO: Coordinating Entropy via Gradient-Preserving Clipping Policy Optimization in
536 Reinforcement Learning. In *ACL*, 2026.

537 Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2
538 edition, 2018.

539 Shenzhi Wang, Le Yu, Chang Gao, Chujie Zheng, Shixuan Liu, Rui Lu, Kai Dang, Xionghui Chen,
540 Jianxin Yang, Zhenru Zhang, Yuqiong Liu, An Yang, Andrew Zhao, Yang Yue, Shiji Song, Bowen
541 Yu, Gao Huang, and Junyang Lin. Beyond the 80/20 Rule: High-Entropy Minority Tokens Drive
542 Effective Reinforcement Learning for LLM Reasoning. In *NeurIPS*, 2025.

543 Yige Xu, Xu Guo, Zhiwei Zeng, and Chunyan Miao. SoftCoT: Soft Chain-of-Thought for Efficient
544 Reasoning with LLMs. In *ACL*, 2025.

545 An Yang, Beichen Zhang, Binyuan Hui, Bofei Gao, Bowen Yu, Chengpeng Li, Dayiheng Liu,
546 Jianhong Tu, Jingren Zhou, Junyang Lin, Keming Lu, Mingfeng Xue, Runji Lin, Tianyu Liu,
547 Xingzhang Ren, and Zhenru Zhang. Qwen2.5-Math Technical Report: Toward Mathematical
548 Expert Model via Self-Improvement. *arXiv:2409.12122*, 2024.

549 Wengao Ye, Yan Liang, and Lianlei Shan. Thinking on the Fly: Test-Time Reasoning Enhancement
550 via Latent Thought Policy Optimization. In *ICLR*, 2026.

551 Qiying Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan, Xiaochen Zuo, Yu Yue, Weinan Dai, Tiantian
552 Fan, Gaohong Liu, Juncai Liu, Lingjun Liu, Xin Liu, Haibin Lin, Zhiqi Lin, Bole Ma, Guangming
553 Sheng, Yuxuan Tong, Chi Zhang, Mofan Zhang, Ru Zhang, Wang Zhang, Hang Zhu, Jinhua Zhu,
554 Jiaze Chen, Jiangjie Chen, Chengyi Wang, Hongli Yu, Yuxuan Song, Xiangpeng Wei, Hao Zhou,
555 Jingjing Liu, Wei-Ying Ma, Ya-Qin Zhang, Lin Yan, Yonghui Wu, and Mingxuan Wang. DAPO:
556 An Open-Source LLM Reinforcement Learning System at Scale. In *NeurIPS*, 2025.

557 Weihao Zeng, Yuzhen Huang, Qian Liu, Wei Liu, Keqing He, Zejun Ma, and Junxian He. SimpleRL-
558 Zoo: Investigating and Taming Zero Reinforcement Learning for Open Base Models in the Wild.
559 *arXiv:2503.18892*, 2025.

560 Guibin Zhang, Muxin Fu, and Shuicheng Yan. MemGen: Weaving Generative Latent Memory for
561 Self-Evolving Agents. In *ICLR*, 2026.

562 A Experimental Setup and Full Accuracy Breakdown

563 Table 6 provides the full per-position accuracy breakdown (Prefix, Infix, Suffix) corresponding to
564 the best-across-positions results summarized in Table 1 in the main text. Table 7 lists the number of
565 RSP tokens L used for each model–benchmark pair, selected from $\{10, 15, 20\}$. All experiments are
566 conducted on a single NVIDIA A6000 40GB GPU with greedy decoding, a maximum generation
567 length of 3,072 tokens for MATH-500 and GSM8K, and 4,096 tokens for AIME24. Batch size is
568 fixed per model (16 for LLaMA-3.1-8B-Instruct and Qwen2.5-Math-7B variants, 32 for Qwen2.5-
569 Math-1.5B variants).

Table 6: Accuracy (%) across model configurations, injection positions, and benchmarks. Values in parentheses indicate the difference from the baseline. **Bold** denotes the best result and underline denotes the second best for each model–benchmark pair.

Model	Mode	MATH-500	GSM8K	AIME24
<i>Instruct Models</i>				
LLaMA-3.1-8B-Instruct	Baseline	52.20	85.75	<u>6.67</u>
	Prefix	45.00 (−7.2)	76.35 (−9.4)	3.33 (−3.3)
	Infix	49.40 (−2.8)	85.97 (+0.2)	10.00 (+3.3)
	Suffix	<u>49.40</u> (−2.8)	86.73 (+1.0)	10.00 (+3.3)
Qwen2.5-Math-7B-Instruct	Baseline	83.20	95.45	<u>13.33</u>
	Prefix	81.40 (−1.8)	94.92 (−0.5)	13.33 (+0.0)
	Infix	84.20 (+1.0)	95.60 (+0.2)	16.67 (+3.3)
	Suffix	<u>83.40</u> (+0.2)	95.68 (+0.2)	16.67 (+3.3)
Qwen2.5-Math-1.5B-Instruct	Baseline	73.00	<u>85.29</u>	10.00
	Prefix	74.80 (+1.8)	85.29 (+0.0)	<u>13.33</u> (+3.3)
	Infix	75.20 (+2.2)	85.75 (+0.5)	6.67 (−3.3)
	Suffix	74.20 (+1.2)	84.69 (−0.6)	20.00 (+10.0)
<i>Base Models (Raw Text)</i>				
Qwen2.5-Math-7B	Baseline	72.20	84.15	6.67
	Prefix	<u>71.60</u> (−0.6)	84.31 (+0.2)	16.67 (+10.0)
	Infix	63.20 (−9.0)	64.29 (−19.9)	16.67 (+10.0)
	Suffix	55.60 (−16.6)	54.13 (−30.0)	<u>13.33</u> (+6.7)
Qwen2.5-Math-1.5B	Baseline	<u>61.40</u>	77.86	16.67
	Prefix	64.40 (+3.0)	74.30 (−3.6)	10.00 (−6.7)
	Infix	63.20 (+1.8)	73.92 (−3.9)	<u>10.00</u> (−6.7)
	Suffix	54.60 (−6.8)	58.61 (−19.3)	3.33 (−13.3)
<i>Base Models + ChatML (Format Mismatch)</i>				
Qwen2.5-Math-7B	Baseline	52.20	58.30	23.33
	Prefix	59.20 (+7.0)	56.41 (−1.9)	3.33 (−20.0)
	Infix	<u>62.00</u> (+9.8)	69.07 (+10.8)	23.33 (+0.0)
	Suffix	70.40 (+18.2)	75.97 (+17.7)	23.33 (+0.0)
Qwen2.5-Math-1.5B	Baseline	34.40	37.45	<u>13.33</u>
	Prefix	63.40 (+29.0)	67.02 (+29.6)	20.00 (+6.7)
	Infix	39.20 (+4.8)	50.42 (+13.0)	6.67 (−6.7)
	Suffix	<u>51.00</u> (+16.6)	<u>50.64</u> (+13.2)	<u>13.33</u> (+0.0)

Table 7: Number of RSP tokens (L) used for each model and benchmark.

Model	MATH-500	GSM8K	AIME24
LLaMA-3.1-8B-Instruct	15	20	15
Qwen2.5-Math-7B-Instruct	20	20	20
Qwen2.5-Math-1.5B-Instruct	20	10	20
Qwen2.5-Math-7B	10	10	20
Qwen2.5-Math-1.5B	10	10	20
Qwen2.5-Math-1.5B (ChatML)	20	20	10
Qwen2.5-Math-7B (ChatML)	20	20	20

570 A.1 RSP Injection Positions and Prompt Templates

571 Below we show the prompt structure for each injection position across all model types. Blue text
572 indicates RSP embeddings, and purple text indicates special tokens.

573 A.1.1 Prefix

574 RSP embeddings are concatenated before the prompt embeddings. No text modification is applied.

575 ChatML (Qwen Instruct / Base + ChatML).

```
[Random Embeddings ( $L$  tokens)]  
<|im_start|>system  
Please reason step by step, and put your final answer within \boxed{>.  
<|im_start|>user  
{question}<|im_end|>  
<|im_start|>assistant
```

577 LLaMA Chat Template.

```
[Random Embeddings ( $L$  tokens)]  
<|begin_of_text|>  
<|start_header_id|>system<|end_header_id|>  
Cutting Knowledge Date: December 2023  
Today Date: 26 Jul 2024  
Please reason step by step, and put your final answer within \boxed{>.  
<|start_header_id|>user<|end_header_id|>  
{question}<|eot_id|>  
<|start_header_id|>assistant<|end_header_id|>
```

579 Raw Text (Qwen Base).

```
[Random Embeddings ( $L$  tokens)]  
{question}  
Please reason step by step, and put your final answer within \boxed{>.
```

581 A.1.2 Infix

582 A description text and special tokens are inserted into the prompt. The special token positions are
583 then replaced with RSP embeddings at the embedding level.

584 ChatML (Qwen Instruct / Base + ChatML).

```
<|im_start|>system  
Please reason step by step, and put your final answer within \boxed{>.  
<|im_start|>user  
{question}  
  
There are { $L$ } special tokens that contain compressed latent reasoning information that  
might be useful for your reasoning. If these tokens are useful for your case, you can  
use them as reference. If these tokens are not useful for your case, you can ignore  
them and focus back to solving the problem.  
  
Here are the { $L$ } special tokens: <special_token_1>...<special_token_ $L$ >  
<|im_end|>  
<|im_start|>assistant
```

586 LLaMA Chat Template.

```
<|begin_of_text|>  
<|start_header_id|>system<|end_header_id|>  
Cutting Knowledge Date: December 2023  
Today Date: 26 Jul 2024  
Please reason step by step, and put your final answer within \boxed{>.  
<|start_header_id|>user<|end_header_id|>
```

{question}

There are $\{L\}$ special tokens that contain compressed latent reasoning information that might be useful for your reasoning. If these tokens are useful for your case, you can use them as reference. If these tokens are not useful for your case, you can ignore them and focus back to solving the problem.

Here are the $\{L\}$ special tokens:

```
<reserved_special_token_0>...  
<reserved_special_token_L>  
<|eot_id|>  
<|start_header_id|>assistant<|end_header_id|>
```

588

589 Raw Text (Qwen Base).

{question}

There are $\{L\}$ special tokens that contain compressed latent reasoning information that might be useful for your reasoning. If these tokens are useful for your case, you can use them as reference. If these tokens are not useful for your case, you can ignore them and focus back to solving the problem.

Here are the $\{L\}$ special tokens: $\langle \text{special_token_1} \rangle \dots \langle \text{special_token_L} \rangle$

Please reason step by step, and put your final answer within `\boxed{\}`.

590

591 A.1.3 Suffix

592 For chat-templated models, RSP embeddings are inserted immediately before the end-of-turn token.

593 For raw text models, RSP embeddings are concatenated at the end of the prompt.

594 ChatML (Qwen Instruct / Base + ChatML).

```
<|im_start|>system  
Please reason step by step, and put your final answer within \boxed{<|im_end|>  
<|im_start|>user  
{question}[Random Embeddings ( $L$  tokens)]<|im_end|>  
<|im_start|>assistant
```

595

596 LLaMA Chat Template.

```
<|begin_of_text|>  
<|start_header_id|>system<|end_header_id|>  
Cutting Knowledge Date: December 2023  
Today Date: 26 Jul 2024  
Please reason step by step, and put your final answer within \boxed{<|eot_id|>  
<|start_header_id|>user<|end_header_id|>  
{question}[Random Embeddings ( $L$  tokens)]<|eot_id|>  
<|start_header_id|>assistant<|end_header_id|>
```

597

598 Raw Text (Qwen Base).

```
{question}  
Please reason step by step, and put your final answer within \boxed{<|eot_id|>[Random  
Embeddings ( $L$  tokens)]
```

599

600 A.2 Answer Extraction and Grading

601 The final answer is extracted from the model output through a fallback chain. Each step is attempted
602 in order; if extraction fails, the next step is tried.

Answer Extraction Fallback Chain

1. "final answer is \$...\$. I hope" pattern (Minerva format)
2. `\boxed{...}`: last non-empty box is used; empty `\boxed{}` are skipped
3. Text following "the answer is"
4. Text following "final answer is"
5. Last number in the output (regex fallback)

Extracted answers are normalized by removing \$, \left, \right, and unit strings. Grading is performed via symbolic equivalence checking: exact string match, numerical comparison (`math.isclose, rel_tol=1e-4`), and SymPy symbolic simplification.

A.3 Reproduced Prior Work Hyperparameters

All prior methods are reproduced on Qwen2.5-Math-1.5B-Instruct and Qwen2.5-Math-7B-Instruct, evaluated with greedy decoding (temperature = 0) and our unified answer extraction pipeline.

TTSV. Prefix length 20, trained for 20 epochs with AdamW optimizer, batch size 2, gradient accumulation 8 steps, and linear learning rate schedule. Learning rate is 1e-3 for Qwen-1.5B and 1e-5 for Qwen-7B.

LTPO. Table 8 reports the per-benchmark hyperparameters.

Table 8: LTPO hyperparameters.

Parameter	GSM8K	MATH-500	AIME24
tokens	8	8	8
steps	20	20	20
top_k	10	10	10
sigma	20	20	5
sigma_decay	0.95	0.95	0.95
lr	1e-2	5e-3	5e-2

SoftCoT. Projection module trained for 10 epochs. Thought tokens: 32 during training, 4 during evaluation. Batch size 4 for GSM8K, 1 for MATH with gradient accumulation 4. The small (assistant) model is Qwen2.5-Math-1.5B-Instruct in both configurations, while the large model is Qwen2.5-Math-1.5B-Instruct (learning rate 2e-5) or Qwen2.5-Math-7B-Instruct (learning rate 5e-6). For MATH-500 and AIME24 evaluation, we use the projection module trained on GSM8K, as it yields higher accuracy than the one trained on MATH.

A.4 Full Entropy Curves

Figure 6 shows the full normalized (0–100%) generation trajectories for entropy, top-1 probability, and varentropy, and Table 9 reports the corresponding scalar statistics including overall means, first-5% means, and the average number of generated tokens per problem. All methods converge to similar levels after the initial generation stage, confirming that the distributional changes induced by RSP are localized to the early reasoning phase.

Table 9: Per-step statistics for Qwen2.5-Math-1.5B-Instruct on MATH-500 across the full generation and the first 5% of generation steps, together with the average number of generated tokens per problem. Top-1 Prob (overall) is reported as a weighted average over the first-10%/middle-last-10% segment means with weights 0.1/0.8/0.1. Extension of Figure 3 (main text) and Figure 6.

Metric	Baseline	Prefix	Infix	Suffix	LTPO	SoftCoT	TTSV
Tokens (mean)	575.7	558.4	549.9	573.3	592.3	594.4	577.4
Entropy (first 5%)	0.1332	0.1366	0.1402	0.1941	0.1662	0.4218	0.1359
Entropy (overall)	0.0871	0.0881	0.0899	0.1049	0.0909	0.1059	0.0864
Top-1 Prob (first 5%)	0.9535	0.9523	0.9525	0.9373	0.9437	0.9156	0.9534
Top-1 Prob (overall)	0.9684	0.9681	0.9678	0.9647	0.9683	0.9651	0.9685
Varentropy (first 5%)	0.2181	0.2207	0.2463	0.4010	0.2953	2.2234	0.2174
Varentropy (overall)	0.1353	0.1375	0.1422	0.2038	0.1452	0.2404	0.1361

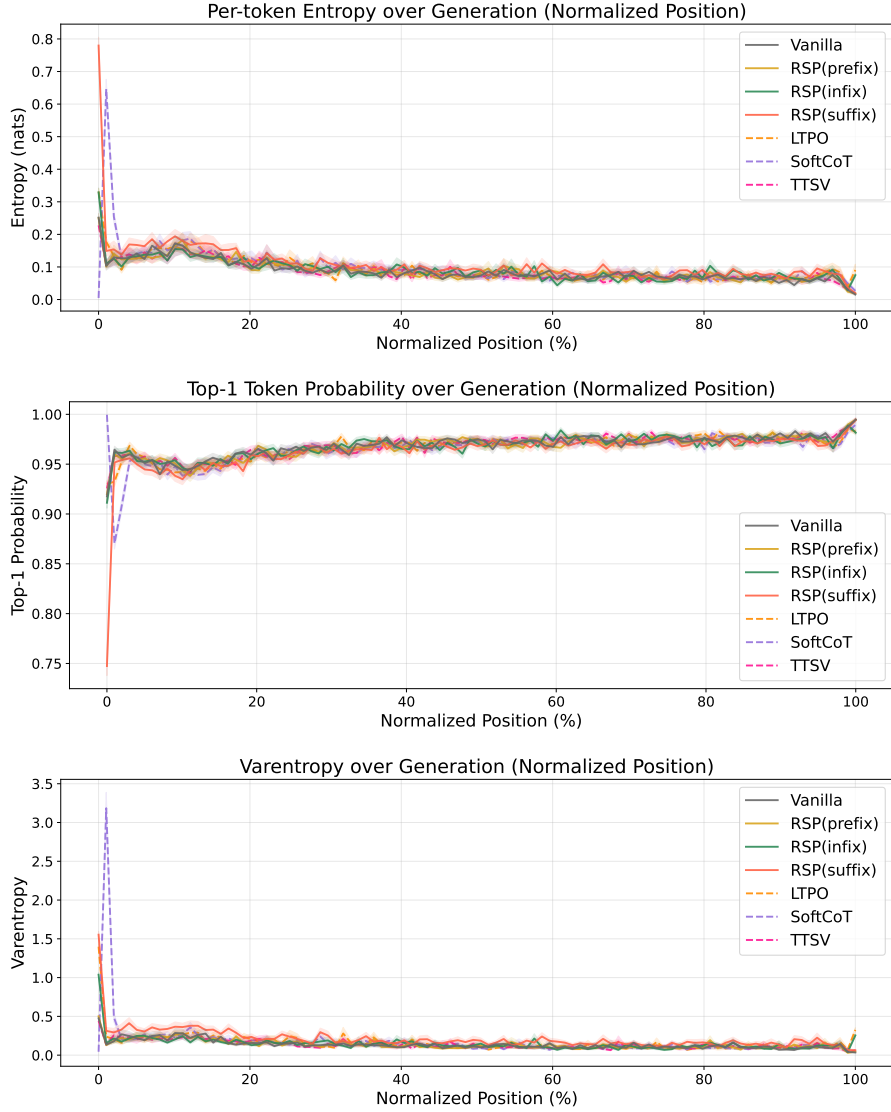


Figure 6: Mean entropy (top), top-1 probability (middle), and varentropy (bottom) over the full normalized generation trajectory (0–100%). Averaged over MATH-500, shading indicates ± 1 SEM. Solid lines: RSP variants and the baseline. Dashed lines: existing soft prompt methods. All methods converge after the initial stage.

626 B Token-Level Distribution Metrics

627 This appendix provides formal definitions for the metrics used in Section 4.4. Numerical results
 628 are reported in Table 3 of the main text. We extract first-token logits via a single forward pass (no
 629 autoregressive generation) and store the top-100 token ids and logit values per problem. Temperature
 630 conditions reuse the baseline logits scaled by $1/\tau$, requiring no separate inference. RSP draws 16
 631 independent seeds per problem (MATH-500), and reported metrics are the mean across the 16 seeds.
 632 Pass@1 accuracy is computed over 16 samples per problem, with the baseline at $\tau=1.0$ reaching
 633 $73.92 \pm 0.78\%$. Let P_{base} denote the baseline next-token distribution at $\tau = 1.0$ and P_{target} the
 634 candidate distribution being compared (e.g., baselines at $\tau=2.0, 3.0$ or RSP at $\tau = 1.0$). All metrics
 635 are computed analytically from saved logits at the first generation step.

636 **Spearman rank correlation.** Spearman’s ρ is the Pearson correlation coefficient between the token
 637 ranks of two distributions:

$$\rho = \text{Pearson}(\text{rank}(P_{\text{target}}), \text{rank}(P_{\text{base}})). \quad (4)$$

638 Because temperature scaling $P^{(\tau)} \propto P^{1/\tau}$ is a strictly monotone transform, the rank order of tokens
 639 is preserved exactly, so $\rho = 1$ for any $\tau > 0$ as a mathematical identity. Any value $\rho < 1$ therefore
 640 indicates rank reorganization beyond what temperature scaling can produce.

641 **Mass outside top- K .** Let $\text{TopK}(P_{\text{base}})$ be the set of K tokens with the highest probability under
 642 P_{base} . The probability mass that the candidate distribution places *outside* this set is

$$\text{MassOutside}_K = 1 - \sum_{t \in \text{TopK}(P_{\text{base}})} P_{\text{target}}(t). \quad (5)$$

643 This directly measures support expansion: how much probability the candidate assigns to tokens that
 644 are not in the baseline’s preferred set. Reported values use $K = 10$.

645 **Jensen–Shannon divergence.** The symmetrized, bounded version of Kullback–Leibler divergence:

$$\text{JS}(P, Q) = \frac{1}{2} \text{KL}(P \parallel M) + \frac{1}{2} \text{KL}(Q \parallel M), \quad M = \frac{1}{2}(P + Q). \quad (6)$$

646 We use base-2 logarithms so that $\text{JS} \in [0, 1]$. Tokens absent from the stored top-100 are assigned a
 647 sentinel logit (-10^9) before softmax, which is negligible for our setting.

C Prompt-Law Properties Used by the Theory

This appendix isolates the only properties of the RSP law used in the main text: centeredness, direction-agnostic covariance under a variance budget, and positive probability on every nonempty open set. We avoid stronger semantic claims because Section 3.4 needs only these geometric and measure-theoretic facts.

C.1 Centered Perturbations Do Not Impose Systematic First-Order Drift

Let Δ_{ij} be the baseline logit gap between tokens i and j , and write the first-order surrogate as

$$\Delta_{ij}(\bar{h}) = \Delta_{ij} + b_{ij}^\top \bar{h}.$$

Proposition 3 (No systematic first-order drift). *If $\mathbb{E}[\bar{h}] = 0$, then*

$$\mathbb{E}[\Delta_{ij}(\bar{h})] = \Delta_{ij}$$

for every token pair (i, j) . If a deterministic prompt h_0 is reused across runs, then

$$\Delta_{ij}(h_0) - \Delta_{ij} = b_{ij}^\top h_0$$

is a fixed directional bias.

Proof. The centered case is immediate from linearity of expectation:

$$\mathbb{E}[\Delta_{ij}(\bar{h})] = \Delta_{ij} + b_{ij}^\top \mathbb{E}[\bar{h}] = \Delta_{ij}.$$

The deterministic case follows by substitution. $\square$

This proposition rules out a run-averaged first-order bias. It does not say that individual prompt draws leave the logits unchanged.

C.2 Proof of Theorem 2

For a unit vector b , the first-order perturbation energy available along b is $\text{Var}_{h \sim D}(b^\top h) = b^\top \Sigma_D b$. The worst-case directional energy is therefore the minimum Rayleigh quotient of Σ_D .

Proof. For any symmetric covariance matrix Σ_D ,

$$\min_{\|b\|=1} b^\top \Sigma_D b = \lambda_{\min}(\Sigma_D).$$

Let the eigenvalues of Σ_D be $\lambda_1, \dots, \lambda_d$. Then

$$\lambda_{\min}(\Sigma_D) \leq \frac{1}{d} \sum_{m=1}^d \lambda_m = \frac{\text{tr}(\Sigma_D)}{d} \leq \frac{\rho^2}{d}.$$

Equality holds if and only if all eigenvalues are equal, i.e., $\Sigma_D = (\rho^2/d)\mathbf{I}_d$. $\square$

The proof uses only covariance. We choose a Gaussian law for RSP because it adds the open-set reachability property proved next.

C.3 Open-Set Reachability of Isotropic Gaussian Prompts

Proposition 4 (Open-set reachability). *Let $\bar{h} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_d)$ with $\sigma > 0$. Then every nonempty open set $U \subseteq \mathbb{R}^d$ satisfies*

$$\Pr(\bar{h} \in U) > 0.$$

Proof. The Gaussian density

$$(2\pi\sigma^2)^{-d/2} \exp\left(-\frac{\|\bar{h}\|^2}{2\sigma^2}\right)$$

is strictly positive at every point of $\mathbb{R}^d$. Every nonempty open set contains a ball of positive Lebesgue measure, and integrating a strictly positive density over that ball gives positive probability. $\square$

This is the only support property used in Proposition 1.

677 D Proofs for the Transformer-Level Mechanism

678 This appendix follows the same order as the main theory section: exploration, annealing, and
679 performance gain. The prompt-law facts used before these proofs are collected in Appendix C.

680 D.1 Exploration: attention decomposition and perturbation size

681 **Proposition 5** (Attention decomposition). *For one attention head at query position i in layer ℓ , let*
682 *$\alpha_{ij}^{(\ell)}$ be the attention weights over $n \geq 1$ unmasked real tokens and $L \geq 1$ random tokens, with finite*
683 *attention logits. Define*

$$w_{r,i}^{(\ell)} := \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)}$$

684 and, for $j \in [n]$,

$$\tilde{\alpha}_{ij}^{(\ell)} := \frac{\alpha_{ij}^{(\ell)}}{1 - w_{r,i}^{(\ell)}}.$$

685 Then

$$\mathbf{o}_i^{(\ell)} = (1 - w_{r,i}^{(\ell)}) \tilde{\mathbf{o}}_i^{(\ell)} + \boldsymbol{\eta}_i^{(\ell)},$$

686 where

$$\tilde{\mathbf{o}}_i^{(\ell)} := \sum_{j=1}^n \tilde{\alpha}_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} \quad \text{and} \quad \boldsymbol{\eta}_i^{(\ell)} := \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)}.$$

687 *Proof.* The head output is

$$\mathbf{o}_i^{(\ell)} = \sum_{j=1}^n \alpha_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} + \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)}.$$

688 By the softmax positivity and the existence of at least one unmasked real token, $\sum_{j=1}^n \alpha_{ij}^{(\ell)} =$
689 $1 - w_{r,i}^{(\ell)} > 0$, so

$$\sum_{j=1}^n \alpha_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} = (1 - w_{r,i}^{(\ell)}) \sum_{j=1}^n \frac{\alpha_{ij}^{(\ell)}}{1 - w_{r,i}^{(\ell)}} \mathbf{v}_j^{(\ell)} = (1 - w_{r,i}^{(\ell)}) \tilde{\mathbf{o}}_i^{(\ell)}.$$

690 Substituting this identity gives the decomposition. □

691 **Corollary 1** (Magnitude scales with random-token attention). *For every realization of the random*
692 *values,*

$$\|\boldsymbol{\eta}_i^{(\ell)}\| \leq w_{r,i}^{(\ell)} \max_{j \in [L]} \|\mathbf{v}_{r_j}^{(\ell)}\|.$$

693 *Proof.* By the triangle inequality,

$$\|\boldsymbol{\eta}_i^{(\ell)}\| = \left\| \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)} \right\| \leq \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)} \|\mathbf{v}_{r_j}^{(\ell)}\| \leq w_{r,i}^{(\ell)} \max_{j \in [L]} \|\mathbf{v}_{r_j}^{(\ell)}\|.$$

694 □

695 Corollary 1 is the only quantitative fact needed in the main text. No independence assumption
696 between attention weights and random keys is required. Once $w_{r,i}^{(\ell)}$ is small, the random contribution
697 must be small as well.

698 D.2 Corollaries of Theorem 1 on cache growth

699 The fixed-gap decay bound yields the corollaries below.

700 **Corollary 2** (Sequence-wise annealing of the fixed-gap envelope). *For fixed L and $\Delta_i^{(\ell)}$, the upper*
 701 *bound*

$$f(n) = \frac{L}{L + n \exp(\Delta_i^{(\ell)} / \sqrt{d})}$$

702 *is strictly decreasing in n and tends to 0 as $n \rightarrow \infty$. Cache growth alone therefore narrows the*
 703 *largest RSP attention mass compatible with that gap.*

704 During autoregressive decoding $\Delta_i^{(\ell)}$ also moves because queries, keys, and hidden states co-evolve.
 705 The accurate reading is not “the realized mass decreases at every step” but “the envelope compatible
 706 with a fixed gap shrinks as the cache grows.”

707 *Proof.* Differentiate $f(n)$ with respect to n :

$$f'(n) = -\frac{L \exp(\Delta_i^{(\ell)} / \sqrt{d})}{\left(L + n \exp(\Delta_i^{(\ell)} / \sqrt{d})\right)^2} < 0.$$

708 Thus $f(n)$ is strictly decreasing, and its denominator diverges linearly in n while its numerator is
 709 fixed, so $f(n) \rightarrow 0$. $\square$

710 **Corollary 3** (Vanishing random contribution under annealing). *If $n \rightarrow \infty$ while L and $\Delta_i^{(\ell)}$ remain*
 711 *fixed, $w_{r,i}^{(\ell)} \rightarrow 0$, and for every realization of the random values*

$$\|\boldsymbol{\eta}_i^{(\ell)}\| \leq w_{r,i}^{(\ell)} \max_{j \in [L]} \|\mathbf{v}_{r_j}^{(\ell)}\| \rightarrow 0.$$

712 *Proof.* $w_{r,i}^{(\ell)} \rightarrow 0$ by Corollary 2. Apply Corollary 1 to bound $\|\boldsymbol{\eta}_i^{(\ell)}\|$. The maximum norm is finite
 713 for any realization of finitely many sampled values. $\square$

714 D.3 Performance gain: from local admission to Pass@N

715 The main text §3.4 uses a first-order surrogate to state two claims: a baseline-excluded token can
 716 enter the local top- K set, and independent reruns can turn such local openings into Pass@ N gains.
 717 The proofs below use only one support condition on the centered prompt law D :

$$(S1) \quad D(U) > 0 \quad \text{for every nonempty open set } U \subseteq \mathbb{R}^d.$$

718 Proposition 4 shows that the isotropic Gaussian law of §3.4 satisfies (S1).

719 D.3.1 Autoregressive Formulation

720 In autoregressive decoding, the joint distribution over a sequence $y_{1:T}$ factorizes as

$$\Pr(y_{1:T} \mid x) = \prod_{t=1}^T p(y_t \mid x, y_{<t}),$$

721 where each step is governed by a prefix-conditioned distribution. Any perturbation introduced by
 722 RSP affects generation through a sequence of local changes to $p(y_t \mid x, y_{<t})$, rather than a single
 723 global distribution.

724 D.3.2 Local Linearization of Logits under RSP

725 Throughout this subsection, $\bar{h} \in \mathbb{R}^d$ denotes *one* centered prompt vector from the RSP definition
 726 (Eq. (1) in §3.1), treated as a per-token surrogate. The actual implementation uses a length- L prompt
 727 $\mathbf{H} = (h_1, \dots, h_L)$ with L independent draws; the arguments below isolate how one local perturbation

728 can shift the logits. We model the perturbed logits as a function $z_i(\bar{h})$, assuming local smoothness in
 729 a neighborhood of $\bar{h} = 0$:

$$z_i(\bar{h}) = z_i(0) + \nabla_h z_i(0)^\top \bar{h} + r_i(\bar{h}),$$

730 where $r_i(\bar{h}) = O(\|\bar{h}\|^2)$. Defining $a_i := \nabla_h z_i(0)$, we obtain the local affine surrogate

$$z_i^{\text{lin}}(\bar{h}) := z_i + a_i^\top \bar{h}.$$

731 This approximation is used as a first-order surrogate for branch opening, not as an exact global model
 732 of the transformer logits.

733 D.3.3 Proof of Proposition 1

734 Let

$$\mathcal{G}_{i,K} := \{\bar{h} \in \mathbb{R}^d : \#\{j \neq i : z_j^{\text{lin}}(\bar{h}) > z_i^{\text{lin}}(\bar{h})\} \leq K - 1\}.$$

735 *Proof.* Assume $\mathcal{G}_{i,K}$ contains a nonempty open set U . By (S1), $D(U) > 0$. Since $U \subseteq \mathcal{G}_{i,K}$,

$$\Pr_{\bar{h}}(\bar{h} \in \mathcal{G}_{i,K}) \geq D(U) > 0.$$

736 This is exactly the event that token i enters the surrogate top- K set.

737 A useful sufficient condition is the existence of some $\bar{h}_0$ at which token i strictly outranks all but
 738 at most $K - 1$ other tokens in the affine surrogate. By continuity, a neighborhood of $\bar{h}_0$ is then
 739 contained in $\mathcal{G}_{i,K}$. $\square$

740 D.3.4 Proof of Proposition 2

741 This proposition does not derive $p_{\min}(x) > 0$ from the mechanism. It only converts a task-side lower
 742 bound and independence into Pass@ N growth.

743 *Proof.* Let A_n be the event that the n -th independent RSP run reaches a correct trajectory on problem
 744 x , and write $p_n(x) := \Pr(A_n) \geq p_{\min}(x)$ as in (M1). Under (M2),

$$\Pr\left(\bigcap_{n=1}^N A_n^c\right) = \prod_{n=1}^N (1 - p_n(x)) \leq \prod_{n=1}^N (1 - p_{\min}(x)) = (1 - p_{\min}(x))^N.$$

745 Taking complements gives

$$\Pr(\text{Pass@}N \text{ on } x) = \Pr\left(\bigcup_{n=1}^N A_n\right) \geq 1 - (1 - p_{\min}(x))^N.$$

746 The bound $1 - x \leq e^{-x}$ for $x \in [0, 1]$ then yields

$$1 - (1 - p_{\min}(x))^N \geq 1 - e^{-N p_{\min}(x)}.$$

747 $\square$

E Attention Mass Analysis

This appendix formalizes the per-token attention mass reported in Figure 2 and the exclusion criteria applied to the reasoning span and the layer axis. For each of the 500 MATH-500 problems, a fresh RSP $\mathbf{H} \in \mathbb{R}^{L \times d}$ with $L = 10$ is sampled independently, so the reported heatmaps average over both problem variability and RSP-vector variability.

E.1 Per-Token Attention Mass

For each problem, we measure attention on the concatenated sequence [prompt; $\mathbf{H}$; generation], where the generation is the trajectory produced by the same model under matching RSP injection. Let $\alpha_{i,j}^{(\ell,h)}$ denote the post-softmax attention weight at layer ℓ , head h , from query position i to key position j , satisfying $\sum_j \alpha_{i,j}^{(\ell,h)} = 1$ under the causal mask. Let Q and R denote the index sets of question and RSP tokens with sizes $|Q|$ and $|R| = L$, and let H be the number of heads. The per-token attention mass that query i directs to each group at layer ℓ is

$$\text{PTM}_Q[\ell, i] = \frac{1}{|Q|H} \sum_{h=1}^H \sum_{j \in Q} \alpha_{i,j}^{(\ell,h)}, \quad \text{PTM}_R[\ell, i] = \frac{1}{|R|H} \sum_{h=1}^H \sum_{j \in R} \alpha_{i,j}^{(\ell,h)}. \quad (7)$$

Normalization by $|Q|$ and $|R|$ controls for group size, so both quantities express how much attention a single question (resp. RSP) token receives on average under the same softmax denominator. Question tokens thereby serve as a size-matched baseline that absorbs sequence-length effects common to both groups.

E.2 Heatmap Aggregation

We partition the layer indices and the reasoning position indices each into five equal-size bins $\{B_L^{(b)}\}_{b=1}^5$ and $\{B_P^{(b)}\}_{b=1}^5$. For sample s and target group $\bullet \in \{Q, R\}$, the value in cell (b_L, b_P) is

$$M_\bullet^{(s)}[b_L, b_P] = \frac{1}{|B_L^{(b_L)}| \cdot |B_P^{(b_P)}|} \sum_{\ell \in B_L^{(b_L)}} \sum_{i \in B_P^{(b_P)}} \text{PTM}_\bullet[\ell, i]. \quad (8)$$

Figure 2 reports the sample average of Eq. (8) over the 500 valid samples.

E.3 Excluding the `\boxed{\dots}` Span

The reasoning position range terminates strictly before the final `\boxed{\dots}` span. Chain-of-thought outputs decompose into a *text* (content) component and a *pattern* (template) component that contribute independently to downstream performance [Madaan and Yazdanbakhsh, 2022]. The `\boxed{\dots}` wrapper is a canonical pattern: a short, stereotyped span whose role is to complete the answer format rather than to extend reasoning. Including it would therefore inject a template-driven attention signature unrelated to reasoning dynamics.

E.4 Excluding the Final Two Layers

The layer axis further excludes the last two transformer layers. This choice is motivated by a standard late-layer effect rather than by any model-specific tuning decision.

Output-projection specialization. Late-layer hidden states lie in an approximately linear relationship with vocabulary logits and therefore function as a progressive linear readout rather than as representation-transforming blocks [Belrose et al., 2023]. Consistently, late-layer feed-forward modules have been characterized as output-distribution shaping mechanisms [Geva et al., 2021]. Attention in these layers therefore reflects output formation rather than reasoning computation.

Excluding the final two layers keeps the heatmap focused on reasoning-phase dynamics instead of readout-phase dynamics. This exclusion is not load-bearing for the main claim: the sequence-direction attenuation emphasized in the main text is also visible without it, and Theorem 1 concerns sequence-direction attenuation rather than layer-wise monotonicity.

787 E.5 Additional Suffix Heatmaps

788 Figure 7 reports the same per-token RSP attention analysis under suffix injection for Qwen2.5-Math-
 789 1.5B-Instruct and LLaMA-3.1-8B-Instruct. For the late-layer reasons discussed above, the pattern
 790 does not generalize uniformly across every cell; nevertheless, the overall trend is consistent across
 791 both models: question-token attention varies broadly across layers, whereas RSP attention shows the
 792 layer-wise and sequence-wise decay predicted by Theorem 1.

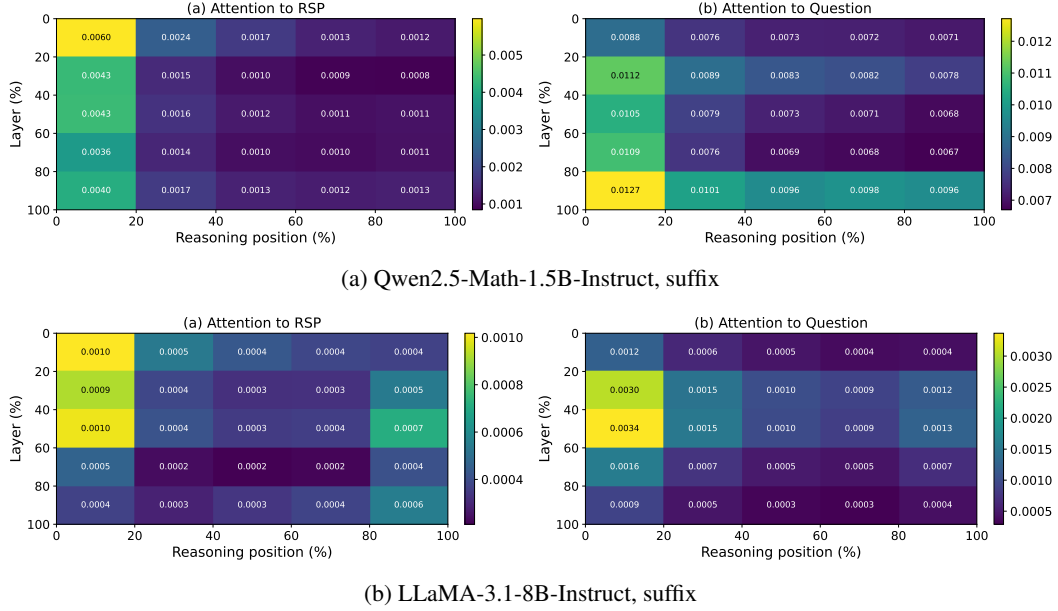


Figure 7: Per-token RSP attention mass under suffix injection for the remaining two models (500 MATH-500 samples each). Axes and preprocessing match Figure 2: x-axis is reasoning position binned into five quantiles, y-axis is layer depth binned into five quantiles (top = shallow), color encodes the 500-sample mean per-token attention assigned to RSP tokens, and tokens inside `\boxed{}` spans and the last two layers are excluded.

F DAPO Implementation

This appendix complements Section 5 with the DAPO loss modifications relative to GRPO, the DAPO + RSP implementation, and full per-step results.

F.1 From GRPO to DAPO

The vanilla GRPO objective [Shao et al., 2024] for a group of G rollouts $\{o_i\}_{i=1}^G$ with rewards $\{R_i\}_{i=1}^G$ is

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \left(\min(r_{i,t} \hat{A}_{i,t}, \text{clip}(r_{i,t}, 1-\varepsilon, 1+\varepsilon) \hat{A}_{i,t}) - \beta D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}) \right) \right], \quad (9)$$

with importance ratio $r_{i,t} = \pi_\theta(o_{i,t} | q, o_{i,<t}) / \pi_{\theta_{\text{old}}}(o_{i,t} | q, o_{i,<t})$, group-relative advantage $\hat{A}_{i,t} = (R_i - \text{mean}(\{R_i\})) / \text{std}(\{R_i\})$, clipping range ε , and KL penalty βD_{KL} against the reference π_{ref} .

We build on the SimpleRL-Zoo [Zeng et al., 2025] training pipeline and implement DAPO on top of VeRL [Sheng et al., 2024]. DAPO modifies Eq. (9) in four ways: (i) token-level loss aggregation $1 / \sum_i |o_i|$ instead of $1 / |o_i|$, (ii) asymmetric clipping with $(\varepsilon_{\text{low}}, \varepsilon_{\text{high}}) = (0.2, 0.28)$, (iii) dual clipping safeguard $\max(L_{\text{clipped}}, c\hat{A})$ with $c = 10$, and (iv) KL terms removed. The resulting DAPO objective is

$$\mathcal{J}_{\text{DAPO}}(\theta) = \mathbb{E} \left[\frac{1}{\sum_i |o_i|} \sum_{i,t} \min(r_{i,t} \hat{A}_{i,t}, \text{clip}(r_{i,t}, 1-\varepsilon_{\text{low}}, 1+\varepsilon_{\text{high}}) \hat{A}_{i,t}) \right]. \quad (10)$$

F.2 DAPO + RSP Implementation

For DAPO + RSP, we additionally inject a group-specific RSP $\mathbf{H}_g \in \mathbb{R}^{L \times d}$ ($L = 20$) into each rollout. $\mathbf{H}_g$ is generated deterministically from a per-step group seed and injected via a forward hook; it is not a trainable parameter, so the learning signal is the policy adapting to arbitrary $\mathbf{H}_g$ rather than a learned prompt. Including $\mathbf{H}_g$ in the log-probabilities at both rollout and update time keeps the update on-policy with respect to the RSP-conditioned distribution (avoiding off-policy mismatch). With $G = 8$ and $n = 8$, each group contains a single response, so each rollout effectively sees a different $\mathbf{H}_g$.

F.3 Data, Batch, and Hyperparameters

Table 10: DAPO / DAPO + RSP training setup on Qwen2.5-Math-7B.

Field	Value
Train data	MATH Level 3–5 subset (simplelr_math_35, ~8,500 prompts)
Prompt template	qwen-boxed
Train batch size	1024
PPO mini-batch size	256 (4 PPO updates per rollout)
Rollouts per prompt G	8
Max prompt / response length	2,028 / 2,048
Rollout sampling	temperature 1.0, top- p 1.0, top- k 50
$(\varepsilon_{\text{low}}, \varepsilon_{\text{high}})$	(0.2, 0.28)
Dual clip c	10.0
Loss aggregation	token-mean ($1 / \sum_i o_i $)
KL terms	disabled
Entropy coefficient	0
Optimizer	AdamW, learning rate 5×10^{-7} (constant), no warmup
Total training	~10 epochs, ≥ 80 optimization steps
Save / eval frequency	every 10 steps
RSP (DAPO + RSP only)	suffix injection, $L = 20$, freshly drawn per rollout
Hardware	4× NVIDIA B200 GPUs
Wall-clock training time	~12 hours per run

815 **F.4 Evaluation Protocol**

816 Each saved checkpoint is converted to HuggingFace format and evaluated using the SimpleRL-
 817 Zoo `eval_math_nodes.sh` pipeline, which selects sampling parameters per benchmark to reduce
 818 variance on the smaller competition sets:

Table 11: Per-benchmark evaluation protocol used by SimpleRL-Zoo `sh/eval.sh`.

Benchmark	# Problems	Temperature	n_{sampling}	Metric
AIME 2024	30	1.0	32	Avg@32
AMC 2023	40	1.0	32	Avg@32
MATH-500	500	0.0	1	accuracy (mean@1)
GSM8K	1,319	0.0	1	accuracy
OlympiadBench	675	0.0	1	accuracy
Minerva Math	272	0.0	1	accuracy
GaoKao 2023-EN	385	0.0	1	accuracy

819 The maximum number of generated tokens is 16,000 across all benchmarks. The reported average is
 820 the unweighted mean over the five benchmarks (GSM8K, MATH-500, College Math, Minerva Math,
 821 AIME 2024).

822 **F.5 Per-Step Average Accuracy**

Table 12: Five-benchmark average accuracy (%) at every checkpoint, computed over GSM8K, MATH-500, College Math, Minerva Math, and AIME 2024. **Bold** marks each method’s peak.

Step	DAPO	DAPO + RSP
0	30.06	30.06
10	46.40	46.22
20	49.02	49.54
30	49.58	50.72
40	50.62	52.06
50	50.88	52.40
60	51.84	52.96
70	52.52	53.86
80	53.20	54.04
90	52.52	54.36
100	50.54	53.64

823 G Broader Impacts

824 This work provides empirical and theoretical analysis of how random embedding injection affects
825 the reasoning behavior of large language models. The contributions are primarily foundational,
826 with potential downstream implications for reinforcement learning-based post-training of reasoning
827 models such as GRPO.

828 **Positive impacts.** RSP-induced trajectory diversity could improve the sample efficiency of RL-
829 based LLM post-training by providing richer learning signals within group-relative advantage estima-
830 tion. This has the potential to reduce compute costs associated with alignment and reasoning-oriented
831 fine-tuning, making such methods more accessible to research groups with limited resources.

832 **Potential concerns.** Methods that expand the effective output support of LLMs may increase the
833 reachability of low-probability tokens, which includes both desirable alternative reasoning paths
834 and potentially unsafe or off-distribution outputs. Practitioners applying RSP in production systems
835 should combine it with existing safety filtering and alignment mechanisms rather than treating it as a
836 standalone diversity source.

837 **Limitations on impact scope.** Since this work focuses on analysis rather than deploying a new
838 model or dataset, the direct societal impact is limited. The broader implications described above are
839 contingent on future work integrating RSP into applied training pipelines.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and §1 state four scoped claims, each tied to a section: (i) RSP’s contribution at each attention layer is captured by a single scalar with a fixed-gap upper-envelope cache-growth bound (Theorem 1, §3.3); (ii) under a local affine surrogate, isotropic re-sampling assigns positive probability to a top- K entry region unattainable by rank-preserving temperature sampling (Theorem 1, §3.4); (iii) RSP raises early-token entropy and stabilizes later, with Pass@ N gains under independent re-sampling (§4); and (iv) the diversity transfers to RL training when combined with DAPO (§5). We do not claim a theorem that directly explains the format-mismatch recovery (Table 1); the wrong-attractor hypothesis there is explicitly informal.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Limitations are noted in several places rather than in a dedicated section: (a) Proposition 2 requires the task-side assumption $p_{\min}(x) > 0$, flagged in §3.4 and §6; (b) the +18–29 pp recovery on the format-mismatch rows of Table 1 is empirically observed but not formally derived from any theorem in §3, with the wrong-attractor account stated as a hypothesis to be quantified in future work (§4.2); (c) Table 1 reports the best-position accuracy across {Prefix, Infix, Suffix}, and per-position variance is large with several cases below baseline (Appendix A), so position is treated as a hyperparameter rather than something we justify principally; (d) DAPO + RSP results (§5) use a single training seed under a reduced DAPO configuration that omits Dynamic Sampling and overlong reward shaping; (e) main accuracy comparisons are single-seed without bootstrap or significance tests (item 7). Future work in §6 addresses training-time comparisons, domain expansion, and principled position selection.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution

is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.

- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: Theorem 1 (cache-growth attenuation) carries an inline proof in §3.3. Proposition 2, Theorem 1, and Proposition 2 are stated in §3.4 with proofs in Appendix D. Each statement explicitly lists its assumptions: finite logits and $n, L \geq 1$ for Theorem 1; locally affine surrogate around $\bar{h} = 0$ for Theorem 1; and $p_{\min}(x) > 0$ together with mutual independence of the N runs for Proposition 2.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: §4.1 and Appendix A specify benchmarks (MATH-500, GSM8K, AIME 2024 in the main table; further sets used for the DAPO study), models (LLaMA-3.1-8B-Instruct and Qwen2.5-Math 1.5B/7B variants), the RSP injection grid (Prefix/Infix/Suffix; $L \in \{10, 15, 20\}$), greedy decoding, and the answer extraction pipeline. Appendix F provides the full DAPO + RSP training recipe (batch of 1,024 prompts, $G = 8$ rollouts per prompt at $\tau = 1.0$, four PPO mini-batch updates of 256 prompts each, 100 training steps, suffix RSP $L = 20$ during rollouts only).

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.

- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility, and authors should be aware that reproducibility may be limited in such cases.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: All datasets and base models used in the paper are publicly available and cited (MATH-500, GSM8K, AIME 2024, AMC 2023, OlympiadBench, Minerva Math, GaoKao 2023-EN, College Math; Qwen2.5-Math 1.5B/7B and LLaMA-3.1-8B-Instruct). RSP itself is described to a level of detail sufficient for re-implementation (Eq. (1) and §4.1). However, our RSP injection harness and DAPO + RSP training pipeline are not yet released; we will release an anonymized implementation alongside the camera-ready submission.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

1000 Question: Does the paper specify all the training and test details (e.g., data splits, hyperpa-
1001 rameters, how they were chosen, type of optimizer) necessary to understand the results?

1002 Answer: [Yes]

1003 Justification: §4.1 and Appendix A list all benchmarks, models, RSP injection settings
1004 (positions, lengths, per-row L choice in Appendix Table 7), maximum generation length
1005 per benchmark, and decoding strategy. Appendix F provides the full DAPO + RSP training
1006 settings, including optimizer, batch composition, mini-batch updates, asymmetric clipping
1007 range $(\epsilon_{\text{low}}, \epsilon_{\text{high}}) = (0.2, 0.28)$, dual clipping safeguard, and RSP injection protocol during
1008 rollouts.

1009 Guidelines:

- 1010 • The answer [N/A] means that the paper does not include experiments.
- 1011 • The experimental setting should be presented in the core of the paper to a level of detail
1012 that is necessary to appreciate the results and make sense of them.
- 1013 • The full details can be provided either with the code, in appendix, or as supplemental
1014 material.

1015 7. Experiment statistical significance

1016 Question: Does the paper report error bars suitably and correctly defined or other appropriate
1017 information about the statistical significance of the experiments?

1018 Answer: [No]

1019 Justification: Pass@ N experiments use $N = 16$ samples per problem (§4.4), Table 3 reports
1020 mean±standard deviation over 16 RSP seeds, and AIME 2024 is scored by Avg@32 to
1021 reduce variance. However, Table 1 (Table 1) reports single-run accuracies without bootstrap
1022 confidence intervals or paired tests, and the DAPO + RSP training in §5 uses a single training
1023 seed. Multi-seed evaluation, particularly on small competition sets (AIME 2024, AMC
1024 2023) where variance is highest, is left for future work.

1025 Guidelines:

- 1026 • The answer [N/A] means that the paper does not include experiments.
- 1027 • The authors should answer [Yes] if the results are accompanied by error bars, confidence
1028 intervals, or statistical significance tests, at least for the experiments that support the
1029 main claims of the paper.
- 1030 • The factors of variability that the error bars are capturing should be clearly stated (for
1031 example, train/test split, initialization, random drawing of some parameter, or overall
1032 run with given experimental conditions).
- 1033 • The method for calculating the error bars should be explained (closed form formula,
1034 call to a library function, bootstrap, etc.)
- 1035 • The assumptions made should be given (e.g., Normally distributed errors).
- 1036 • It should be clear whether the error bar is the standard deviation or the standard error
1037 of the mean.
- 1038 • It is OK to report 1-sigma error bars, but one should state it. The authors should
1039 preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis
1040 of Normality of errors is not verified.
- 1041 • For asymmetric distributions, the authors should be careful not to show in tables or
1042 figures symmetric error bars that would yield results that are out of range (e.g., negative
1043 error rates).
- 1044 • If error bars are reported in tables or plots, the authors should explain in the text how
1045 they were calculated and reference the corresponding figures or tables in the text.

1046 8. Experiments compute resources

1047 Question: For each experiment, does the paper provide sufficient information on the com-
1048 puter resources (type of compute workers, memory, time of execution) needed to reproduce
1049 the experiments?

1050 Answer: [Yes]

1051 Justification: Inference experiments (Tables 1, 2; Figure 3) run on a single NVIDIA A6000
1052 40GB GPU with greedy decoding (Appendix A). DAPO + RSP training (§5) runs on 4×
1053 NVIDIA B200 GPUs for approximately 12 hours per run (Appendix F, hardware row of the
1054 configuration table).

1055 Guidelines:

- 1056 • The answer [N/A] means that the paper does not include experiments.
- 1057 • The paper should indicate the type of compute workers CPU or GPU, internal cluster,
1058 or cloud provider, including relevant memory and storage.
- 1059 • The paper should provide the amount of compute required for each of the individual
1060 experimental runs as well as estimate the total compute.
- 1061 • The paper should disclose whether the full research project required more compute
1062 than the experiments reported in the paper (e.g., preliminary or failed experiments that
1063 didn’t make it into the paper).

1064 9. Code of ethics

1065 Question: Does the research conducted in the paper conform, in every respect, with the
1066 NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines?>

1067 Answer: [Yes]

1068 Justification: The work uses publicly available math reasoning benchmarks and pretrained
1069 language models, involves no human subjects or personal data, and does not introduce a
1070 release that bypasses standard model-safety considerations. We have reviewed the NeurIPS
1071 Code of Ethics and identify no conflicts.

1072 Guidelines:

- 1073 • The answer [N/A] means that the authors have not reviewed the NeurIPS Code of
1074 Ethics.
- 1075 • If the authors answer [No], they should explain the special circumstances that require a
1076 deviation from the Code of Ethics.
- 1077 • The authors should make sure to preserve anonymity (e.g., if there is a special consid-
1078 eration due to laws or regulations in their jurisdiction).

1079 10. Broader impacts

1080 Question: Does the paper discuss both potential positive societal impacts and negative
1081 societal impacts of the work performed?

1082 Answer: [Yes]

1083 Justification: Appendix G discusses both directions: positive impact via more sample-
1084 efficient RL post-training of reasoning LLMs, and a potential downside that broadening the
1085 effective output support also increases reachability of undesirable trajectories, which we
1086 argue should keep RSP coupled with existing safety filtering rather than replacing it.

1087 Guidelines:

- 1088 • The answer [N/A] means that there is no societal impact of the work performed.
- 1089 • If the authors answer [N/A] or [No], they should explain why their work has no societal
1090 impact or why the paper does not address societal impact.
- 1091 • Examples of negative societal impacts include potential malicious or unintended uses
1092 (e.g., disinformation, generating fake profiles, surveillance), fairness considerations
1093 (e.g., deployment of technologies that could make decisions that unfairly impact specific
1094 groups), privacy considerations, and security considerations.
- 1095 • The conference expects that many papers will be foundational research and not tied
1096 to particular applications, let alone deployments. However, if there is a direct path to
1097 any negative applications, the authors should point it out. For example, it is legitimate
1098 to point out that an improvement in the quality of generative models could be used to
1099 generate Deepfakes for disinformation. On the other hand, it is not needed to point out
1100 that a generic algorithm for optimizing neural networks could enable people to train
1101 models that generate Deepfakes faster.

- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The paper does not release new pretrained models, generative systems, or scraped datasets. RSP is a generation-time perturbation applied to existing pretrained models and introduces no separate artifact requiring safeguards.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: All datasets (MATH-500 [Lightman et al., 2024], GSM8K [Cobbe et al., 2021], AIME 2024, AMC 2023, OlympiadBench [He et al., 2024], Minerva Math [Lewkowycz et al., 2022], GaoKao 2023-EN, College Math) and pretrained models (Qwen2.5-Math [Yang et al., 2024], LLaMA-3.1 [Grattafiori et al., 2024]) are publicly available and cited at first use. The training pipeline builds on SimpleRL-Zoo [Zeng et al., 2025] and VeRL [Sheng et al., 2024], also cited. Each asset is used under its original publicly stated terms.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.

- If this information is not available online, the authors are encouraged to reach out to the asset’s creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [N/A]

Justification: The paper does not release new datasets or pretrained models. The implementation code planned for camera-ready release (item 5) will be accompanied by configuration files and a README; no derived dataset or model checkpoint is part of the submission.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The paper involves no crowdsourcing and no research with human subjects. All evaluations use static, publicly available math reasoning benchmarks.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: As noted in item 14, the paper involves no human subjects, so IRB review is not applicable.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.

1206 • For initial submissions, do not include any information that would break anonymity (if
1207 applicable), such as the institution conducting the review.

1208 **16. Declaration of LLM usage**

1209 Question: Does the paper describe the usage of LLMs if it is an important, original, or
1210 non-standard component of the core methods in this research? Note that if the LLM is used
1211 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
1212 scientific rigor, or originality of the research, declaration is not required.

1213 Answer: [N/A]

1214 Justification: Pretrained LLMs (Qwen2.5-Math, LLaMA-3.1) appear in the paper as the
1215 experimental subject of analysis, not as a non-standard component of the methods themselves.
1216 The methods (RSP definition, attention decomposition, DAPO + RSP training) do not rely
1217 on auxiliary LLMs in any non-standard or originality-affecting role.

1218 Guidelines:

1219 • The answer [N/A] means that the core method development in this research does not
1220 involve LLMs as any important, original, or non-standard components.

1221 • Please refer to our LLM policy in the NeurIPS handbook for what should or should not
1222 be described.